or

$$\mu = \frac{n+1}{2}$$

where we have used the famous identity $1+2+\ldots+n = n(n+1)/2$.

Now, cancelling out the terms and we have the required **moment-generating function**:

$$M(t) = (1-\beta t)^{-\alpha} \quad for\ t < \frac{1}{\beta}$$

Cumulative Distribution Function

The formula for the **cumulative distribution function** of the normal distribution is

$$F(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right] \quad \text{for } -\infty < x < \infty,$$

where

erf(z) is the "error function" defined by

$$\operatorname{erf}(z) \equiv \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\,dt.$$

Normal Distribution

Shm

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

where μ is the mean of the distribution and $\sigma > 0$ is the standard deviati

We say that X has the $N(\mu, \sigma^2)$ distribution. Qut

Long derive Mean

Topic No. 170

Derivation of the mean and a variance of the Normal Distribution

Therefore, E(X

$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2}$

Area under the curve

The first integral prese
curve with zero mean an
The integral being an od

Thus, E(X)= μ i.e. μ is

Hence we have the interesting property that the **mean and standard deviation** of the exponential distribution are equal.

This, in turn, means that **the Coefficient of Variation of the Exponential distribution is always equal to 1.**

Note:

PMF; Probability Mass Function
PDF; Probability Density Function
CDF; Cumulative Distribution Function
MGF; Moment Generating Function
CGF; Cumulant Generating Function

Formal Definition of Independence in the case of Discrete Random Variables

Let the random variables X_1 and X_2 have the joint pmf $p(x_1, x_2)$ and the marginal pmfs $p_1(x_1)$ and $p_2(x_2)$ respectively.

The random variables X_1 and X_2 are said to be independent **if and only if**

$$p(x_1, x_2) = p_1(x_1)\, p_1(x_2).$$

Theorem:

If X and Y are random variables with joint distribution function F and marginal distribution functions F_X and F_Y, then X and Y are independent if and only if

$$F(x, y) = F_X(x) F_Y(y)$$

For all $(x, y) \in R^2$.

the 'X_1X_2 floor'

ni Sivarcale nodes distribute

What is meant by the 'X_1X_2 floor'?

the three-dimensional space for example the room in which y
g right now.

cus on the floor of the room.

dge' of the floor will be called the X_1 axis

her 'edge' of the floor will be called the X_2 axis

as such, the floor will be called the 'X_1X_2 floor'

ally it is the X_1X_2-plane)

Derivation of the MGF of a Binomial distribution

Imp S.Q

mgf of a binomial distribution is easily obtained as follows:

$$M(t)=\sum_{x} e^{tx} p(x)=\sum_{x=0}^{n} e^{tx}\binom{n}{x} p^{x}(1-p)^{n-x}$$

$$=\sum_{x=0}^{n}\binom{n}{x}\left(e^{t} p\right)^{x}(1-p)^{n-x}$$

$$=\left[(1-p)+p e^{t}\right]^{n}$$

for all real values of t.

Now, the **mean** μ and **variance** σ^2 of X computed from $M(t)$.

M.C.Q

We know that $\mu=M'(0)$

and

$\therefore$ First we use binomial

Had the true distribution, the binomial, been used instead of the approximation,

$$P(X=1)=\binom{20}{1}(.08)^{1}(.92)^{19}=0.3282$$

For Poisson.

illustrate the use of the Poisson approximation for the nomial, the probability of obtaining exactly one ective tire from a sample of 20 using the formula is culated as follows:

$$(X=1)=\frac{e^{-(20)(.08)}\left[(20)(.08)\right]^{1}}{1!}=\frac{e^{-1.6}\left[1.6\right]^{1}}{1!}=0.3230$$

long

Topic No. 150

Continuous Uniform Distribution
(Rectangular Distribution)
(PDF, CDF and Shape of the distribution)

distribution $b(x;n, p)$ where $n \to \infty$, $p \to 0$, and the product np remains **constant**, we proceed:

long

3:35 **Example:**

SQ

Suppose 8% of the tires manufactured at a particular plant are defective. To illustrate the use of the Poisson approximation for the binomial, the probability of obtaining exactly one defective tire from a sample of 20 using equation (5.18) is calculated as follows:

$$P(X = 1) = \frac{e^{-(20)(.08)}\left[(20)(.08)\right]^1}{1!} = \frac{e^{-1.6}\left[1.6\right]^1}{1!} = 0.3230$$

Scanned with CamScanner

1) First we use binomial

Had the true distribution, the binomial, been used instead of the approximation,

$$P(X=1)=\binom{20}{1}(.08)^1(.92)^{19}=0.3282$$

Long

For Poisson

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$$(X=1)=\frac{e^{-(20)(.08)}\left[(20)(.08)\right]^1}{1!}=\frac{e^{-1.6}[1.6]^1}{1!}=0.3230$$

long

Topic No. 150

Continuous Uniform Distribution
(Rectangular Distribution)
(PDF, CDF and Shape of the distribution)

Marginal distribution of X:

$$f_X(x) = \int_x^1 2dy = 2.[y]_x^1 = 2(1-x),\ 0 < x < 1$$

2M

and

Marginal distribution of Y:

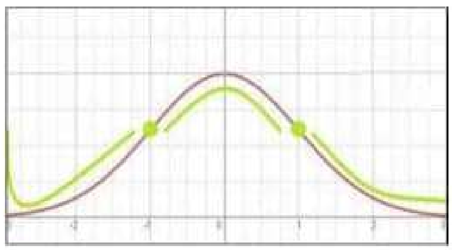
$$f_Y(y) = \int_0^y 2dx = 2.[x]_0^y = 2(y-0) = 2y,\ 0 < y < 1$$

The normal curve has

Points of Inflection

which are equidistant from the mean

Definition: By a 'point of inflection' we mean a point at which the **concavity** of a curve changes,



From calculus, we know that such a point is obtained by solving the equation

$$f''(x) = 0$$

3M

Taking the first derivative of the PDF of the normal distribution, we have

$$f'(x) = \frac{d}{dx}\left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}\right) = \frac{1}{\sigma\sqrt{2\pi}} \frac{d}{dx}\left(e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}\right)$$

$$= \frac{1}{\sigma\sqrt{2\pi}}\left\{e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \frac{d}{dx}\left(-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2\right)\right\}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}\left[\left(-\frac{1}{2}\right).2.\left(\frac{x-\mu}{\sigma}\right)\left(\frac{1}{\sigma}\right)\right]$$

$$= \frac{1}{\ldots} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}\left[-\left(\frac{x-\mu}{\ldots}\right)\right] = -\frac{1}{\ldots}(x-\mu).e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$$

The **variance** of the Multinomial distribution

$$\mathrm{Var}(X_i) = np_i(1 - p_i)$$

2M

The **covariance** of the Multinomial distribution

The mgf of a binomial distribution is easily obtained as follows:

$$M(t) = \sum_{x} e^{tx} p(x) = \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^{x} (1-p)^{n-x}$$

$$= \sum_{x=0}^{n} \binom{n}{x} (e^{t} p)^{x} (1-p)^{n-x}$$

$$= \left[(1-p) + pe^{t}\right]^{n}$$

for all real values of t.

Long

Now, the **mean** μ and **variance** σ^2 of X may be computed from $M(t)$.

We know that $\mu = M'(0)$

and

$\sigma^2 = M''(0) - \mu^2$

(PMF and shape of the distribution)

5M

Geometric Distribution
(PMF and shape of the distribution)

Assume Bernoulli trials — that is,

(1) there are two possible outcomes,
(2) the trials are independent, and

(3) p, the probability of success, remains the same from trial to trial.

Geometric Distribution
(PMF and shape of the distribution)

(4) n, the number of trials is fixed in advance

and

the binomial distribution is given by

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-1}, \; x = 0,1,2,3,\ldots,n$$

When n=1,

there are two possible outcomes,

and p, the probability of success, is the only parameter of the Bernoulli distribution given by

$$p_X(x) = \binom{1}{x} p^x (1-p)^{1-x}, \; x = 0,1$$

or

$$p_X(x) = p^x (1-p)^{1-x}, \; x = 0,1$$

Long 3-Mrs

Similarly

Q.

$$E(X^2|y)=\int_0^1 x^2 . f(x|y)dx=\int_0^y x^2 . \frac{1}{y}dx=\frac{1}{y}\left(\frac{x^3}{3}\right|_0^y$$

Expected Value

$$=\frac{1}{y}\left(\frac{y^3}{3}-0\right)=\frac{1}{y}.\frac{y^3}{3}=\frac{y^2}{3}, \quad 0<y<1$$

Lóng

Example:

If the probability that a person will <u>believe</u> a rumor about the retirement of a certain politician is 0.25, what is the probability that

the sixth person to hear the rumor will be the first to believe it.

Long

We derive the formula for the mean of the discrete random variable $X \sim U(1,n)$:

$P_X = \frac{1}{n}$

$$\mu = \sum_{x=1}^{n} x p_X(x) = \sum_{x=1}^{n} x . \frac{1}{n}$$

$$= \frac{1}{n}(1+2+...+n) = \frac{1}{n} \times \frac{n(n+1)}{2}$$

$- \sum x$

3 - Marks

write 3 - properties

Hypergeometric Distribution

The following conditions characterize the hypergeometric distribution:

1. The result of each draw (the elements of the population being sampled) can be classified into one of two mutually exclusive categories
   (e.g. Pass/Fail or Employed/Unemployed).

5-Marks

Example:

If the probability that a person will believe a rumor about the retirement of a certain politician is 0.25, what is the probability that

the sixth person to hear the rumor will be the first to believe it.

2 — SQ

Poisson Distribution
(PMF and shape of the distribution)

e = 2.718

Consider the function $p(x)$ defined by

$$p(x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^{x}}{x!} & x = 0,1,2,\ldots,\ \lambda > 0 \\ 0 & \text{elsewhere,} \end{cases} \qquad (1)$$

λ is +ve ①

where $m > 0$.

5-Marks

Expectation and Variance

If $X \sim U(a, b)$, then:

$$E(X) = \frac{(a+b)}{2}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

...oof of Variance:

... definition,

Var(X ...

Now,

$E(X^2)$ ...

$= \frac{b^3 - \ldots}{3(b - \ldots}$

Cumulative distribution Function

- The formula for the cumulative distribution function of the beta distribution is also called the **incomplete β function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha,\beta) = \frac{\int_0^x t^{\alpha-1}(1-t)^{\beta-1}}{B(\alpha,\beta)}, \qquad 0 \le x \le 1; \alpha,\beta > 0$$

where B is the beta function defined above.

Short / Long

Mode:

- If $\alpha >$ 1

interio

is

If α o

Long

Derivation of the Variance

If X has a gamma distribution with parameters α and β,

The variance of X is

$$\sigma_X^2 = E\left(X^2\right) - \left[E\left(X\right)\right]^2$$

Topic No. 177

Short

Cumulant Generating Function

of the $N(\mu, \sigma^2)$ distribution

and

its utilization for finding its cumulants

3 Marks

We know that, the **mgf** of Normal Distribution is as

$$M(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}, \qquad \text{for} \quad -\infty < t < \infty.$$

Solution:

We know that, in a sequence **of independent Bernoulli(p) trials**, if the random variable **X** denotes the number of the trial at which the r^{th} **success** occurs, then then the **probability mass function** of X is:

$$P(X=x)=\binom{x-1}{r-1}p^r(1-p)^{x-r},\ x=r, r+1, r+2,...$$

Here, **we do have** a sequence of Bernoulli trials as — whenever we will dig an exploratory oil well, either we will or will not strike oil.

We can assume that the trials are **independent**,

and, **due to the independence** of the trials, we can assume that p, the probability of success, remains the **same** from trial to trial.

If we regard Striking oil as Success, then p**=0.20** and r**=2**

and, as such, the PMF of the negative binomial distribution will be given by

$$P(X=x)=\binom{x-1}{r-1}p^r(1-p)^{x-r},\ x=r, r+1, r+2,...$$

putting the values

$$P(X=x)=\binom{x-1}{2-1}(0.2)^2(0.8)^{x-2},\ x=2,3,4,...$$

$$\text{or } P(X=x)=\binom{x-1}{1}(0.2)^2(0.8)^{x-2},\ x=2,3,4,...$$

$$\text{or } P(X=x)=(x-1)(0.2)^2(0.8)^{x-2},\ x=2,3,4,...$$

vip

To find the required probability, we need to find $P(X=5)$.

$$P(X=x)=(x-1)(0.2)^2(0.8)^{x-2}$$
$$\Rightarrow P(X=5)=(5-1)(0.2)^2(0.8)^{5-2}$$
$$=4(0.2)^2(0.8)^3$$
$$=0.082$$
$$=8.2\%$$

Example:

- A representative from an ice-cream producing company randomly stops people on a randomly selected street in the city of Lahore until he finds a person who actually purchased and tried the latest product of this company meaning a new kind of ice-cream they have just now brought into the market.

- Let p, the probability that, for any one of the persons stopped, he succeeds in finding such a person, who has actually tried that product equals 0.20.

- And, let X denote the number of people he stops until he finds his first success.

- How many people should we **expect** the marketing representative needs to stop before he finds one who actually purchased and tried this ice-cream?

- And, while we're at it, what is the variance?

Solution:

'How many people should we expect' means 'what is the average number).

For the geometric random variable X, the average number is given by:

$$\mu = E(X) = \frac{1}{p} = \frac{1}{0.20} = 5$$

So here

Geometric Distribution
(PMF and shape of the distribution)

Assume Bernoulli trials — that is,

(1) there are two possible outcomes,

(2) the trials are independent, and

(3) p, the probability of success, remains the same from trial to trial.

Same

Geometric Distribution
(PMF and shape of the distribution)

(4) n, the number of trials is fixed in advance

and

the binomial distribution is given by

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-1}, \; x = 0,1,2,3,\ldots,n$$

When n=1,

there are two possible outcomes,

and p, the probability of success, is the only parameter of the Bernoulli distribution given by

$$p_X(x) = \binom{1}{x} p^x (1-p)^{1-x}, \; x = 0,1$$

or

$$p_X(x) = p^x (1-p)^{1-x}, \; x = 0,1$$

Therefore

$$\sum_{x} p(x) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = e^{-\lambda+\lambda} = e^0 = 1$$

That is , $p(x)$ **satisfies** the conditions of being a pmf of a discrete type of random variable.

A random variable that has a pmf of the form $p(x)$ given by eq.(1) is said to have a **Poisson distribution** with parameter λ.

and any such $p(x)$ is called a Poisson pmf with parameter m.

An interesting property:

It is easy to prove that the **mean** and **variance** of the Poisson distribution given by (1) are **both equal to λ.**

Hypergeometric Distribution

The following conditions characterize the hypergeometric distribution:

1. The result of each draw (the elements of the population being sampled) can be classified into one of two mutually exclusive categories

   (e.g. Pass/Fail or Employed/Unemployed).

2. The successive draws are **not independent** of each other.

3. The number of draws is **fixed** in advance.

4. The probability of a success **changes** on each draw, as each draw decreases the population (*sampling without replacement* from a finite population).

Topic No. 139

Derivation of
the Mean and Variance of
the Hypergeometric Distribution

Two important properties of the hypergeometric probability distribution are given here.

The mean & variance of the Hypergeometric distribution are as follows:

$$\text{Mean} = E(X) = n\frac{K}{N}$$

&

$$\text{Variance} = n\frac{K}{N}\frac{(N-K)}{N}\frac{N-n}{N-1}$$

Or --- in other words --- the mean and variance of the hypergeometric probability distribution are

$$\mu = np \text{ and } \sigma^2 = npq\left(\frac{N-n}{N-1}\right),$$

$$\text{where } p = \frac{k}{N} \text{ and } q = \frac{N-k}{N}.$$

$1-p$

Derivation of the
MGF
of a
Binomial distribution

S.Q

The mgf of a binomial distribution is easily obtained as follows:

$$
\begin{aligned}
M(t) = \sum_{x} e^{tx} p(x) &= \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^{x} (1-p)^{n-x} \\
&= \sum_{x=0}^{n} \binom{n}{x} \left(e^{t} p\right)^{x} (1-p)^{n-x} \\
&= \left[(1-p) + pe^{t}\right]^{n}
\end{aligned}
$$

for all real values of t.

Now, the **mean** μ and **variance** σ^2 of X may be computed from $M(t)$.

We know that $\mu = M'(0)$

and

$\sigma^2 = M''(0) - \mu^2$

Mean and Variance of the Negative Binomial Distribution

The mean of the distribution is given by

$$E(X) = \frac{r(1-p)}{p}$$

and the Variance is

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$$

MGF of the Negative Binomial Distribution

The moment generating function of the distribution is given by

$$M(t) = \left(\frac{1-p}{1-pe^t}\right)^r \quad for \ \ t < -\log p$$

This can be used to find the mean, variance and higher moments of the distribution.

A Special Case of the Negative Binomial Distribution: <u>Geometric Distribution</u>

In a sequence of independent Bernoulli(p) trials,
let the random variable X denote the number of the trial at which the ***1**^{st}* **success** occurs.

Negative Binomial Distribution

(PMF, Shape of the distribution, Mean and Variance)

Consider **Bernoulli trials** — that is,

(1) there are two possible outcomes,

(2) the trials are independent,

and

(3) p, the probability of success, remains the same from trial to trial.

In a sequence of such trials, let the random variable X denote the number of the trial at which the r^{th} success occurs, where r is a fixed integer.

Then the **probability mass function** of X is:

$$P(X=x)=\binom{x-1}{r-1}p^r(1-p)^{x-r},\ x=r,r+1,r+2,\ldots$$

and we say that X has a **negative binomial(r,p)** distribution.

let x denote the no. of heads if we toss a fair coin once.

$\Rightarrow$ X = no. of heads

جب ہم کوائن کو ٹاس کریں گے تو یا تو ہمارے
پاس Tail آئے گی یا پھر Head. ہم جانتے ہیں
کہ Head کے آنے کی probability کیا ہوگی۔
تو Head کے آنے کی Proba— تو half ہی ہوگی۔
اور ٹیل کے آنے کی بھی ہاف ہوگی۔

when Tail occur

$X = 0$

when Head occur

$X = 1$

So

sample space = 2

$$P = \frac{\text{event}}{\text{sample space}}$$

$$P_1 = \frac{1}{2}$$

$$P_2 = \frac{1}{2}$$

Geometric Distribution
(PMF and shape of the distribution)

(4) n, the number of trials is fixed in advance

and

the binomial distribution is given by

$$p_X(x)=\binom{n}{x}p^x(1-p)^{n-1},\ x=0,1,2,3,\ldots,n$$

When n=1,

there are two possible outcomes,

and p, the probability of success, is the only parameter of the Bernoulli distribution given by

$$p_X(x)=\binom{1}{x}p^x(1-p)^{1-x},\ x=0,1$$

or

$$p_X(x)=p^x(1-p)^{1-x},\ x=0,1$$

Bern

If we let X denote the number of trials **until the <u>first</u> success.**

Geom

Then, the **probability mass function** of X is:

$$p_X(x)=(1-p)^{x-1}p,\ x=1,2,3,\ldots$$

Example:

$$\therefore E(X^2) = \frac{nk}{N} + \frac{k(k-1)}{\binom{N}{n}}\binom{N-2}{n-2}$$

$$= \frac{nk}{N} + \frac{k(k-1)(N-2)!}{\left(\frac{N!}{n!(N-n)!}\right)(n-2)!(N-n)!} = \frac{nk}{N} + \frac{k(k-1)(N-2)!n!(N-n)!}{N!(n-2)!(N-n)!}$$

$$= \frac{nk}{N} + \frac{k(k-1)(N-2)!n(n-1)(n-2)!(N-n)!}{N(N-1)(N-2)!(n-2)!(N-n)!} = \frac{nk}{N} + \frac{k(k-1)n(n-1)}{N(N-1)}$$

$$\therefore \sigma^2 = \text{Var}(X) = E(X^2) - \mu^2 = \frac{nk}{N} + \frac{k(k-1)n(n-1)}{N(N-1)} - \left(\frac{nk}{N}\right)^2$$

$$= \frac{nkN(N-1)}{N^2(N-1)} + \frac{(k^2-k)(n^2-n)N}{N^2(N-1)} - \frac{(N-1)n^2k^2}{(N-1)N^2}$$

$$= \frac{nkN(N-1) + (n^2k^2 - kn^2 - nk^2 + nk)N - (N-1)n^2k^2}{N^2(N-1)}$$

$$= \frac{nkN^2 - nkN + n^2k^2N - kn^2N - nk^2N + nkN - n^2k^2N + n^2k^2}{N^2(N-1)}$$

$$= \frac{nkN^2 - kn^2N - nk^2N + n^2k^2}{N^2(N-1)} = \frac{nk(N^2 - nN - kN + nk)}{N^2(N-1)}$$

Now

$$\frac{nk(N^2 - nN - kN + nk)}{N^2(N-1)} = \frac{nk(N-k)}{N.N} \cdot \frac{(N-n)}{N-1}$$

So

$$\text{Var}(X) = npq \cdot \frac{N-n}{N-1}$$

where $p = \dfrac{k}{N}$ and $q = \dfrac{N-k}{N}$.

Virtual University of Pakistan
Probability Distributions

by

Dr. Saleha Naghmi Habibullah

Poisson Distribution
(PMF and shape of the distribution)

Consider the function $p(x)$ defined by

$$p(x)=\begin{cases}\dfrac{e^{-\lambda}\lambda^{x}}{x!} & x=0,1,2,\dots,\ \lambda>0 \\ 0 & elsewhere,\end{cases} \qquad (1)$$

where $m>0$.

Since $\lambda>0$,
therefore $p(x)\geq 0$
and

$$\sum_{x}p(x)=\sum_{x=0}^{\infty}\frac{\lambda^{x}e^{-\lambda}}{x!}=e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^{x}}{x!}$$

Recall that the series

$$1+m+\frac{m^{2}}{2!}+\frac{m^{3}}{3!}+\cdots=\sum_{x=0}^{\infty}\frac{m^{x}}{x!}$$

converges, for all values of m, to e^{m}.

Therefore

$$\sum_{x}p(x)=e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^{x}}{x!}=e^{-\lambda}e^{\lambda}=e^{-\lambda+\lambda}=e^{0}=1$$

That is $p(x)$ satisfies the conditions of being a pmf of a

Binomial Theorem

1004

$$(q+p)^n = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x}$$

Example:

Suppose that a game consists of tossing a fair coin until we obtain the 3rd head.

Then

head is 'success',

p=P(H)=0.5

&

***r*=3**

V i P

$p = \frac{1}{2} = 0.5$

In this case, the negative binomial distribution is given by

$$P(X=x)=\binom{x-1}{3-1}(0.5)^3(1-0.5)^{x-3}\ ,\ x=3,3+1,3+2,\ldots$$

or

$$P(X=x)=\binom{x-1}{2}(0.5)^x\ ,\ x=3,4,5,\ldots$$

Calculation of Probabilities

x	*P(x)*
3	$P(X=3)=\binom{2}{2}(0.5)^3=0.1250$
4	$P(X=4)=\binom{3}{2}(0.5)^4=0.1875$
5	$P(X=5)=\binom{4}{2}(0.5)^5=0.1875$
6	$P(X=6)=\binom{5}{2}(0.5)^6=0.1562$
7	$P(X=7)=\binom{6}{2}(0.5)^7=0.1172$
8	$P(X=8)=\binom{7}{2}(0.5)^8=0.0820$
9	$P(X=9)=\binom{8}{2}(0.5)^9=0.0547$
10	$P(X=10)=\binom{9}{2}(0.5)^{10}=0.0352$
And so on.	

Example:

Suppose that a game consists of tossing a fair coin until we obtain the 3rd head.

Then

head is 'success',

***p=P(H)*=0.5**

&

***r*=3**

vip

In this case, the negative binomial distribution is given by

$$P(X=x)=\binom{x-1}{3-1}(0.5)^3(1-0.5)^{x-3},\ x=3,3+1,3+2,...$$

or

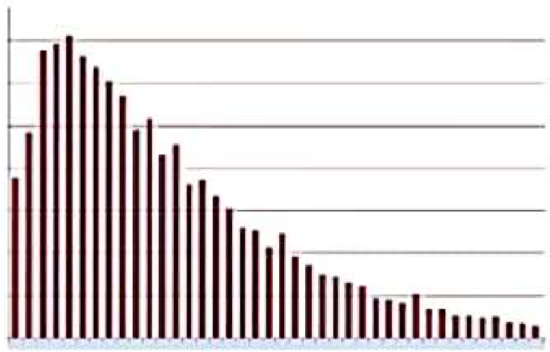
$$P(X=x)=\binom{x-1}{2}(0.5)^x,\ x=3,4,5,...$$

Calculation of Probabilities

x	$P(x)$
3	$P(X=3)=\binom{2}{2}(0.5)^3=0.1250$
4	$P(X=4)=\binom{3}{2}(0.5)^4=0.1875$
5	$P(X=5)=\binom{4}{2}(0.5)^5=0.1875$
6	$P(X=6)=\binom{5}{2}(0.5)^6=0.1562$
7	$P(X=7)=\binom{6}{2}(0.5)^7=0.1172$
8	$P(X=8)=\binom{7}{2}(0.5)^8=0.0820$
9	$P(X=9)=\binom{8}{2}(0.5)^9=0.0547$
10	$P(X=10)=\binom{9}{2}(0.5)^{10}=0.0352$
And so on.	

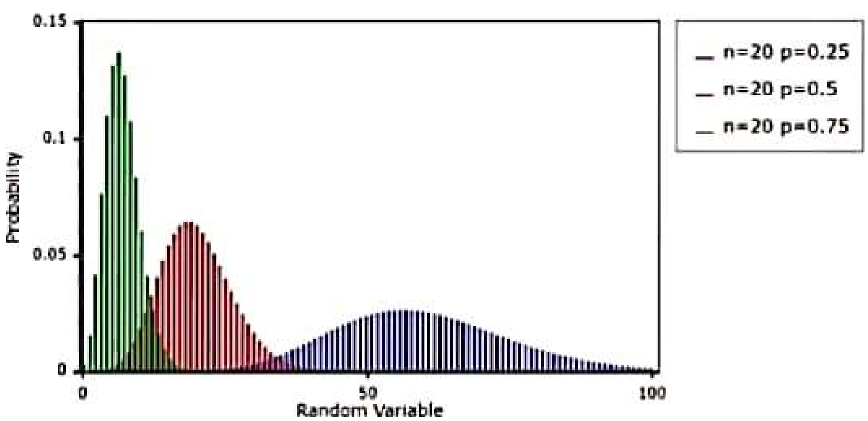
Shape of the distribution

Shape of the distribution



In general,

Shape of the Negative Binomial distribution



It is obvious that the negative binomial distribution is moderately positively skewed and that, as p tends to zero, the shape of the distribution tends to **normality**.

representative needs to stop before he finds one who actually purchased and tried this ice-cream?

- And, while we're at it, what is the variance?

Solution:

'How many people should we expect' means 'what is the average number).

For the geometric random variable X, the average number is given by:

$$\mu = E(X) = \frac{1}{p} = \frac{1}{0.20} = 5$$

So here

$$\mu = E(X) = \frac{1}{0.20} = 5$$

That is, we should expect the marketing representative to have to stop 5 people before he finds one who actually purchased and tried that particular ice-cream.

If this particular experiment of stopping people is repeated again and again, then on the **average** we can expect he would have to stop 5 people and the fifth one will be the one who had tried it.

For the geometric random variable X, the average number is given by:

$$\sigma^2 = Var(X) = \frac{1-p}{p^2}$$

so here

$$\sigma^2 = Var(X) = \frac{1-p}{p^2} = \frac{1-0.20}{0.20^2} = \frac{0.80}{0.04} = 20$$

Example:

The names of 5 men and 5 women are written on slips of paper and placed in a hat. Four names are drawn.

What is the probability that 2 are men and 2 are women?

Let X denote the number of **men.**

size

Then,

N=5+5=10 names to be drawn from;

K=5 & n=4, no.of draws

and here the possible values of x are 0,1,2,3,4, i.e. n)

Hence the hypergeometric distribution is

$$h(x;10,4,5)=\frac{\binom{5}{x}\binom{5}{4-x}}{\binom{10}{4}}$$

and the required probability, i.e. $P(X=2)$ is

$$h(2;10,4,5)=\frac{\binom{5}{2}\binom{5}{4-2}}{\binom{10}{4}}=\frac{\binom{5}{2}\binom{5}{2}}{\binom{10}{4}}=\frac{10}{21}.$$

Shape of the Distribution

https://www.google.com.pk/search?q=hypergeometric+distribution&rlz=1C1CHBD_enPK785PK785&source=lnms&tbm=isch&ved=0ahUKEwjYx9SouNTeAhWBDcAKHZ3hDOkQ_AUIDigB&biw=1280&bih=657#imgrc=6MdjpMk-oHbjKM:

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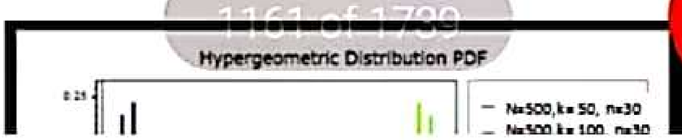

Geometric Distribution
(PMF and shape of the distribution)

(4) n, the number of trials is fixed in advance

and

the binomial distribution is given by

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-1}, \ x = 0,1,2,3,\ldots,n$$

When n=1,

there are two possible outcomes,

and p, the probability of success, is the only parameter of the Bernoulli distribution given by

$$p_X(x) = \binom{1}{x} p^x (1-p)^{1-x}, \ x = 0,1$$

or

$$p_X(x) = p^x (1-p)^{1-x}, \ x = 0,1$$

If we let X denote the number of trials **until the <u>first</u> success.**

Then, the **probability mass function** of X is:

$$p_X(x) = (1-p)^{x-1} p, \ x = 1,2,3,\ldots$$

Example:

Example:

Suppose that a game consists of tossing a fair coin until we obtain the first head.

Then $p=P(H)=0.5$

And, in this case, the geometric distribution is given by

$$p_X(x)=(1-0.5)^{x-1}(0.5)=(0.5)^x,\ x=1,2,3,...$$

or

$$p_X(x)=\frac{1}{2^x},\ x=1,2,3,...$$

Representation in tabular form:

X	$P(x)$
1	$1/2^1=1/2=0.5$
2	$1/2^2=1/4=0.25$
3	$1/2^3=1/8=0.125$
4	$1/2^4=1/16=0.0625$
5	$1/2^5=1/32=0.03125$
6	$1/2^6=1/64=0.015625$

Hypergeometric Distribution

The following conditions characterize the hypergeometric distribution:

1. The result of each draw (the elements of the population being sampled) can be classified into one of two mutually exclusive categories

   (e.g. Pass/Fail or Employed/Unemployed).

2. The successive draws are **not independent** of each other.

3. The number of draws is **fixed** in advance.

4. The probability of a success **changes** on each draw, as each draw decreases the population (*sampling without replacement* from a finite population).

Proof:

By definition

$Var(X)=E(X^2)-[E(X)]^2$, *where*

$$E(X^2)=E[X(X-1)+X]=E(X)+E[X(X-1)]$$

$$=\sum_{x=0}^{\infty}x.\frac{e^{-\mu}\mu^x}{x!}+\sum_{x=0}^{\infty}x(x-1)e^{-\mu}.\frac{\mu^x}{x!}$$

$E(X^2)$

$$=\sum_{x=0}^{\infty}x.\frac{e^{-\mu}\mu^x}{x!}+\sum_{x=0}^{\infty}x(x-1)e^{-\mu}.\frac{\mu^x}{x!}$$

$$=\mu+e^{-\mu}\sum_{x=0}^{\infty}x(x-1).\frac{\mu^x}{x(x-1)(x-2)!}$$

$$=\mu+\mu^2e^{-\mu}\sum_{x=2}^{\infty}\frac{\mu^x\mu^{-2}}{(x-2)!}=\mu+\mu^2e^{-\mu}\sum_{x=2}^{\infty}\frac{\mu^{x-2}}{(x-2)!}$$

(x starts at 2, as the first two terms in the summation are zero.)

Let $y=x-2$, then

$$E(X^2)=\mu+\mu^2e^{-\mu}\sum_{y=0}^{\infty}\frac{\mu^y}{y!} \qquad (y=0,1,2,\ldots)$$

$$=\mu+\mu^2e^{-\mu}e^{\mu}=\mu+\mu^2 \qquad \begin{pmatrix}\text{when } x\to 2, y\to 0,\\ \text{and } x\to\infty, y\to\infty,\end{pmatrix}$$

Hence $Var(X)=\mu+\mu^2-\mu^2=\mu$

Virtual University of Pakistan
Probability Distributions

by

Dr. Saleha Naghmi Habibullah

Edited by

https://newonlinecourses.science.psu.edu/stat414/node/80/

Example

An oil company conducts a geological study that indicates that an exploratory oil well should have a **20% chance** of striking oil.

The company is wanting success **twice** i.e. two oil wells from where they will be able to pull out oil --- and is willing to try upto five times.

What is the probability that the **second success** comes on the **fifth well drilled**?

1. It appears as the limiting form of the binomial distribution under certain conditions.
2. When we apply the Poisson distribution is that real-life situation where we are dealing with Poisson process

Poisson Process

- Whenever a particular event of interest occur randomly over a time-scale, we say that we are dealing with a Poisson process.

Imp

For example, for the Virtual University of Pakistan there is central telephone exchange, and we don't know that when the telephone operator will receive the next call, we from the past experience know that on the average 3 calls are received per minute.

But we do not know that in any one particular span of 1 min exactly how many calls were received? So on

Other than this information, the timings of incoming calls seem to be totally random. Thus, we conclude that the Poisson process might be a good model for the incoming calls at this telephone exchange.

The Poisson Process

The Poisson process is one of the most **widely**-used **counting** processes.

It is usually used in scenarios where we are counting the occurrences of certain events that appear to happen at a certain **rate**, but completely at random (i.e. the events that are happening without any particular structure).

Examples:

When events occur randomly over a time scale:

- From historical data, we know that earthquakes occur in a certain area with a **rate** of 22 per month.
  (Other than this information, the **timings** of earthquakes seem to be **completely random.)**

- From the record of the past six months, we know that, on a particular web server, requests for individual documents occur at the rate of 3 per hour;
  (Other than this information, the timings of the requests seem to be **totally random**.)

A process in which events occur randomly either over a time scale or over a distance scale.

In other words,

a process in which events occur randomly in a specified duration of time or in a specified portion of one-dimensional, two-dimensional or three-dimensional space.

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Examples:

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- From historical data, we know that earthquakes occur in a certain area with a **rate** of 22 per month.
  (Other than this information, the **timings** of earthquakes seem to be **completely random.**)

- From the record of the past six months, we know that, on a particular web server, requests for individual documents occur at the rate of 3 per hour;
  (Other than this information, the timings of the requests seem to be **totally random.**)

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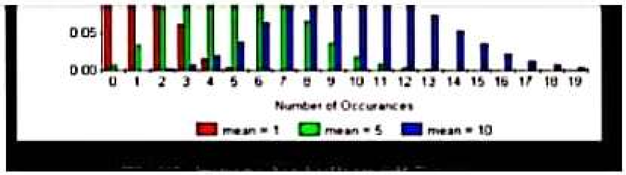
a process in which events occur randomly in a specified duration of time or in a specified portion of one-dimensional, two-dimensional or three-dimensional space.

Examples:



When do we apply this formula?

Poisson distribution arrives in two situations:

1. It appears as the limiting form of the binomial distribution under certain conditions.
2. When we apply the Poisson distribution is that real-life situation where we are dealing with Poisson process

9:46

Mode:

- If $\alpha > 1$ and $\beta > 1$, the peak of the density is in the interior of $[0,1]$ and mode of the Beta distribution is

$$\textbf{mode} = \alpha - 1/\alpha + \beta - 2$$

If α or $\beta < 1$, the mode may be at an edge.

Now, we are required to find the smallest value of n that yields $P(Y \geq 1) > 0.80$.

For this, we proceed as follows:

The condition can be written as $\mathbf{1-(2/3)^{n} > 0.80}$ implying that $0.20 > (2/3)^n$.

Taking log of both sides, we have

$$\ln(0.20) > n \ln(2/3)$$

implying that

$$-1.6094 > n(-0.4055)$$

or

$$-1.6094 > -0.4055n$$

or

$$1.6094 < 0.4055n$$

or

$$n > 1.6094/0.4055 \qquad \text{i.e. } n > 3.97.$$

Thus, we see that $n = 4$ is the solution.

1021

Interpretation:

The probability is greater than 0.80 i.e. 80% that **at least one success** will be obtained in $\mathbf{n = 4}$ independent repetitions of a random experiment for which the probability of success p is equal to 1/3.

Topic No. 127

The sum of m
independent
Binomial random variables
with same value of p
is also Binomial

Theorem:

Let $X_1, X_2, \ldots, X_m$ be **independent** random variables such that X_i has binomial $b(n_i, p)$ distribution, for $i = 1, 2, \ldots, m$.

Let, $$Y = \sum_{i=1}^{m} X_i$$

Then Y has a binomial $b\left(\sum_{i=1}^{m} n_i, p\right)$ distribution.

Proof:

In general, the mgf of a binomial random variable X is

$$M_X(t) = \left(1 - p + pe^t\right)^n.$$

So, in case of m binomial random variables X_i, $i = 1, 2, \ldots, m$, with $p_1 = p_2 = p_3 \ldots\ldots p_m = p$ (i.e. a constant)

the mgf of X_i is $M_{X_i}(t) = \left(1 - p + pe^t\right)^{n_i}$

By **independence** it follows that

$$M_Y(t)=\prod_{i=1}^{m}\left(1-p+pe^t\right)^{n_i}=\left(1-p+pe^t\right)^{\sum_{i=1}^{m}n_i}=\left(q+pe^t\right)^{\sum_{i=1}^{m}n_i}.$$

Hence, $Y=\sum_{i=1}^{m}X_i$ has a binomial $b\left(\sum_{i=1}^{m}n_i, p\right)$ distribution.

Virtual University of Pakistan
Probability Distributions

by

Dr. Saleha Naghmi Habibullah

00:30

Example

The probability that a planted radish seed germinates is 0.80. A gardener plants **nine** seeds.

14

Let X denote the number of radish seeds that successfully germinate.

(i) What is the **average number** of seeds the gardener could expect to germinate?

(ii) What is the **standard deviation** of X?

01:34

(iii) What is the probability that, out of the nine seeds, at least 8 will germinate ?

Derivation of the

MGF

of a

Binomial distribution

125

long

16

00:29

The mgf of a binomial distribution is easily obtained as follows:

$$M(t)=\sum_{x} e^{tx} p(x)=\sum_{x=0}^{n} e^{tx} \binom{n}{x} p^{x}(1-p)^{n-x}$$

$$=\sum_{x=0}^{n} \binom{n}{x} \left(e^{t} p\right)^{x}(1-p)^{n-x}$$

$$=\left[(1-p)+p e^{t}\right]^{n}$$

for all real values of t.

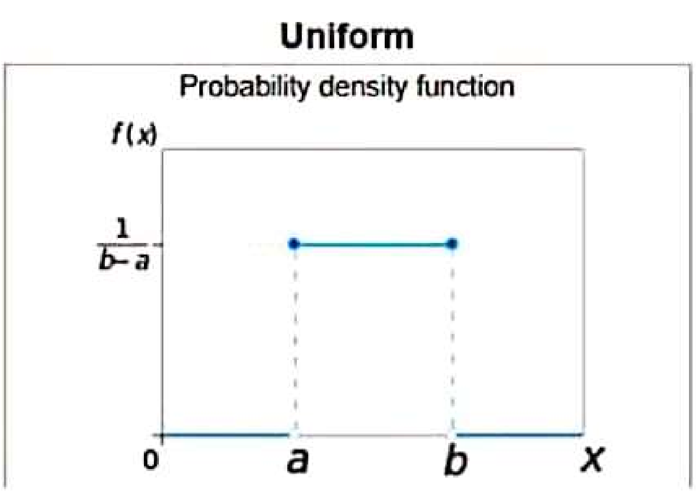
Topic No. 150

Continuous Uniform Distribution
(Rectangular Distribution)
(PDF, CDF and Shape of the distribution)

Probability Distribution Function

The probability density function of the continuous uniform distribution is:

$$f(x)=\begin{cases}\dfrac{1}{b-a} & \text{for } a\le x\le b,\\ 0 & \text{for } x<a \text{ or } x>b\end{cases}$$



Derivation of

the Mean and Variance of the Continuous Uniform distribution

Expectation and Variance

If $X \sim U(a, b)$, then:

$$E(X) = \frac{(a+b)}{2}$$

$$\operatorname{Var}(X) = \frac{(b-a)^2}{12}$$

Proof of Expectation:

The probability density function of the continuous uniform distribution is given by

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a \le x \le b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

Therefore

$$E(X) = \int_{-\infty}^{\infty} x f(x)dx = \int_a^b x\left(\frac{1}{b-a}\right)dx = \frac{1}{b-a}\int_a^b x dx$$

$$= \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)}$$

$$= \frac{b+a}{2} = \frac{a+b}{2}$$

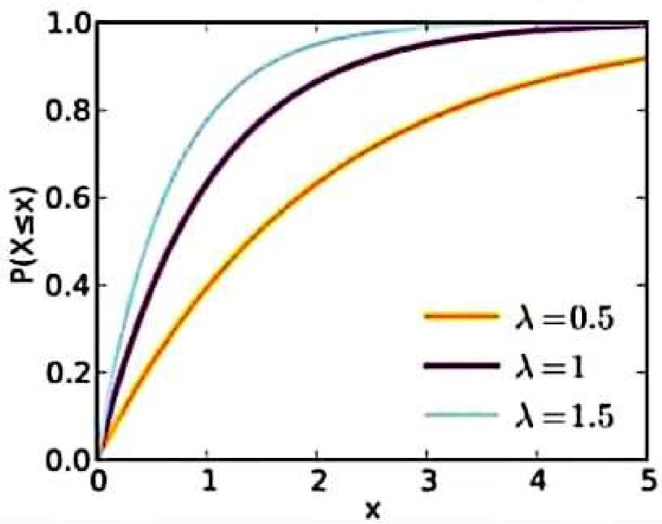
CDF of the **Exponential Distribution** with mean=0.5, 1, 1.5

$$F(x)=1-e^{-x}, \quad 0<x<\infty$$

$$F(x)=1-e^{-\frac{x}{0.5}}, \quad 0<x<\infty$$

$$F(x)=1-e^{-x}, \quad 0<x<\infty$$

$$F(x)=1-e^{-\frac{x}{1.5}}, \quad 0<x<\infty$$



Important Note:

- If we are dealing with Poisson process in which events are occurring randomly over a time scale then the **waiting time** between successive events is distributed according to the exponential distribution.

Derivation of

the Mean of

the Exponential Distribution

A very interesting property:

- The hazard rate of this distribution is constant.

- The **mean and standard deviation** of the exponential distribution are equal.

1. Derivation of the **mean:**

The pdf of the Exponential distribution is given by

$$f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0, \lambda > 0.$$

Therefore,

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx = \int_{0}^{\infty} x.\lambda e^{-\lambda x}\, dx$$

Drivation of Var

Hence, we have

$$E(X^2)=-\left[\frac{x^2}{e^{\lambda x}}\right]_0^\infty+\frac{2}{\lambda}E(X)$$

$$=-[0-0]+\frac{2}{\lambda}E(X)$$

$$=-0+\frac{2}{\lambda}E(X)=\frac{2}{\lambda}E(X)=\frac{2}{\lambda}\left(\frac{1}{\lambda}\right)=\frac{2}{\lambda^2}$$

Hence,

$$\text{Var}(X)=E(X^2)-[E(X)]^2$$

$$=\frac{2}{\lambda^2}-\left(\frac{1}{\lambda}\right)^2=\frac{2}{\lambda^2}-\frac{1}{\lambda^2}=\frac{2-1}{\lambda^2}=\frac{1}{\lambda^2}$$

implying that

$$S.D.(X)=\sigma=\frac{1}{\lambda}.$$

Hence we have the interesting property that the **mean and standard deviation** of the exponential distribution are equal.

This, in turn, means that **the Coefficient of Variation of the Exponential distribution is always equal to 1.**

Explanation:

The coefficient of variation is defined as $CV=\frac{\sigma}{\mu}$.

Hence we have the interesting property that the **mean and standard deviation** of the exponential distribution are equal.

This, in turn, means that **the Coefficient of Variation of the Exponential distribution is always equal to 1.**

Explanation:

The coefficient of variation is defined as $CV = \dfrac{\sigma}{\mu}$.

Hence, for the exponential distribution given by

$f(x) = \lambda e^{-\lambda x}, 0 < x < \infty$

the coefficient of variation is

$$CV = \frac{\sigma}{\mu} = \frac{1/\lambda}{1/\lambda} = \frac{\lambda}{\lambda} = 1.$$

gmp

Example:

The duration of long-distance telephone calls is found to be exponentially distributed with a mean of 3 minutes. What is the probability that a call will last θ

(i) more than 3 minutes, (ii) more than 5 minutes?

x x

Let X be the exponential r.v. with parameter θ.

Then the mean, i.e. $\frac{1}{\theta} = 3$, so that $\theta = \frac{1}{3}$.

and the pdf can be written as

$$f(x) = \int_3^\infty \left(\frac{1}{\theta}\right) e^{-x/\theta} dx$$

Now (i) the probability that a call will last more than 3 minutes, is given by

$$P(X > 3) = \int_3^\infty \left(\frac{1}{3}\right) e^{-x/3} dx = \left[-e^{-x/3}\right]_3^\infty = e^{-1} = 0.3679.$$

$$\int_x^\infty$$

And (ii) the probability that a call will last more than 5 minutes, is given by

$$P(X > 5) = \int_5^\infty \left(\frac{1}{3}\right) e^{-x/3} dx = \left[-e^{-x/3}\right]_5^\infty = e^{-1.7} = 0.1827.$$

1:04 4G 98%

← null.pdf

We know that the probability density function (pdf) of an exponential distribution is

$$f(x;\lambda)=\begin{cases}\lambda e^{-\lambda x} & x\geq 0,\ \lambda>0\\ 0 & x<0.\end{cases}$$

where λ is the only parameter of the distribution.

So to find the MGF, we proceed as follows:

$$M_0(t)=E\left[e^{tX}\right]=\int_0^\infty e^{tx}\lambda e^{-\lambda x}dx=\lambda\int_0^\infty e^{-(\lambda-t)x}dx$$

$$=\lambda\left[\frac{e^{-(\lambda-t)x}}{-(\lambda-t)}\right]_0^\infty=\lambda\left[\frac{-e^{-(\lambda-t)x}}{\lambda-t}\right]_0^\infty$$

Lord

Now, if $t<\lambda$ then $\lambda-t>0$

and we have

$$M_0(t)=\lambda\left[\frac{-e^{-(\lambda-t)\infty}}{\lambda-t}-\frac{-e^{-(\lambda-t)0}}{\lambda-t}\right]=\lambda\left[\frac{-e^{-\infty}}{\lambda-t}-\frac{-e^{0}}{\lambda-t}\right]$$

$$=\lambda\left[\frac{0}{\lambda-t}-\frac{-1}{\lambda-t}\right]=\lambda\left[0+\frac{1}{\lambda-t}\right]=\frac{\lambda}{\lambda-t}$$

Hence, the MGF of the Exponential Distribution is given by

$$M_0(t)=\frac{\lambda}{\lambda-t}\quad\text{for } t<\lambda.$$

Formal definition:

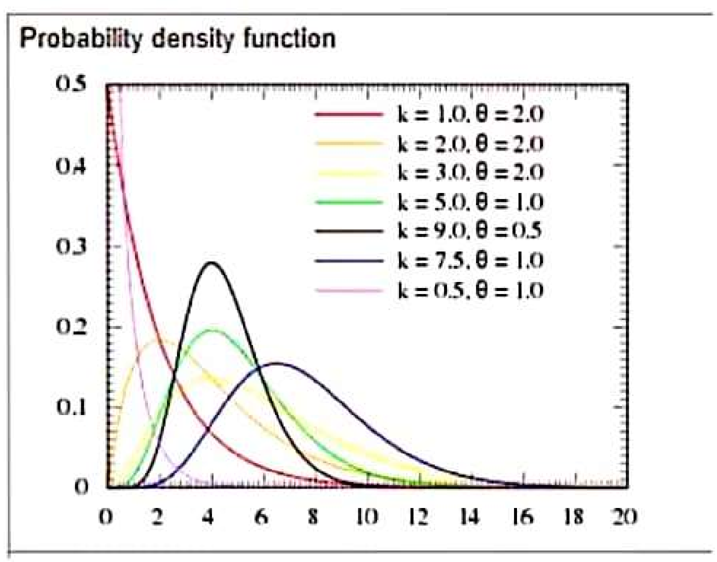
Gamma distribution

$$f(x)=\begin{cases}\dfrac{1}{\Gamma(\alpha)\beta^{\alpha}}x^{\alpha-1}e^{-x/\beta} & 0<x<\infty,\ \alpha>0,\ \beta>0\\ 0 & \text{elsewhere}\end{cases}$$

is said to be the pdf of a gamma distribution with shape parameter α and scale parameter β and we write this as *X* has a $\Gamma(\alpha, \beta)$ distribution.

Shape of the distribution

- From the graph it is obvious that, for any fixed value of the scale parameter β, **as α increases**, the shape of the distribution tends to normality.



First we introduce the **gamma (Γ) function**:

In calculus, the integral

$$\int_0^\infty y^{\alpha-1}e^{-y}\,dy$$

exists for $\alpha > 0$ and that the value of the integral is a positive number.

The integral is called the **gamma function**, and we write

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1}e^{-y}\,dy$$

CDF of the Gamma Distribution

The CDF of the distribution is given by

$$F(x)=\begin{cases}\dfrac{1}{\Gamma(\alpha)}\gamma\left(\alpha,\dfrac{x}{\beta}\right) & \alpha>0,\beta>0\\ 0 & \text{elsewhere}\end{cases}$$

where this $\gamma\left(\alpha,\dfrac{x}{\beta}\right)$ is the incomplete gamma function.

Incomplete gamma function

In mathematics, the **upper** and **lower incomplete gamma functions** are types of special functions which arise as solutions to various mathematical problems such as certain integrals.

ViP

The upper incomplete gamma function is defined as:

$$\Gamma(\alpha,y)=\int_y^{\infty} t^{\alpha-1}e^{-t}dt,$$

whereas the lower incomplete gamma function is defined as:

$$\gamma(\alpha,y)=\int_0^{y} t^{\alpha-1}e^{-t}dt.$$

It is interesting to note that $\gamma(\alpha,y)+\Gamma(\alpha,y)=\Gamma(\alpha)$.

The upper incomplete gamma function is defined as:

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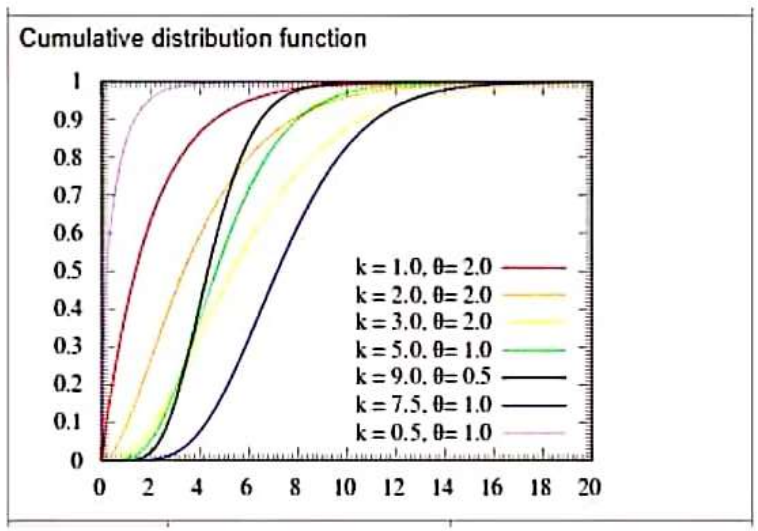
$$\gamma(\alpha, y) = \int_{0}^{y} t^{\alpha-1} e^{-t} dt.$$

It is interesting to note that $\gamma(\alpha, y) + \Gamma(\alpha, y) = \Gamma(\alpha)$.

Shape of the CDF:

We can see that for smaller values of the shape parameter α, i.e. α =0.5 and 1.0, the shape of the CDF appears to be parabolic whereas, for larger values of α i.e. α =2.0, 3.0,5.0 etc , the CDF is S-shaped.



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Hence,

$$E(X)=\alpha\int_0^\infty\left(\frac{x^\alpha e^{-x/\beta}}{\beta^\alpha\,\Gamma(\alpha+1)}\right)dx=\alpha\beta\int_0^\infty\left(\frac{x^\alpha e^{-x/\beta}}{\beta^{\alpha+1}\Gamma(\alpha+1)}\right)dx$$

$$=\alpha\beta\int_0^\infty f(x;\alpha+1,\beta)dx=\alpha\beta(1)=\alpha\beta$$

$\therefore$ The total area under the curve of a probability density function is always equal to 1.

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are defective. To illustrate the use of the Poisson approximation for the binomial, the probability of obtaining exactly one defective tire from a sample of 20 using equation (5.18) is calculated as follows:

$$P(X=1) \equiv \frac{e^{-(20)(.08)}\left[(20)(.08)\right]^1}{1!} = \frac{e^{-1.6}\left[1.6\right]^1}{1!} = 0.3230$$

Had the true distribution, the binomial, been used instead of the approximation,

$$P(X=1) \equiv \binom{20}{1}(.08)^1(.92)^{19} = 0.3282$$

To illustrate the use of the Poisson approximation for the binomial, the probability of obtaining exactly one defective tire from a sample of 20 using the formula is calculated as follows:

$$P(X=1) \equiv \frac{e^{-(20)(.08)}\left[(20)(.08)\right]^1}{1!} = \frac{e^{-1.6}\left[1.6\right]^1}{1!} = 0.3230$$

Cumulative Distribution Function

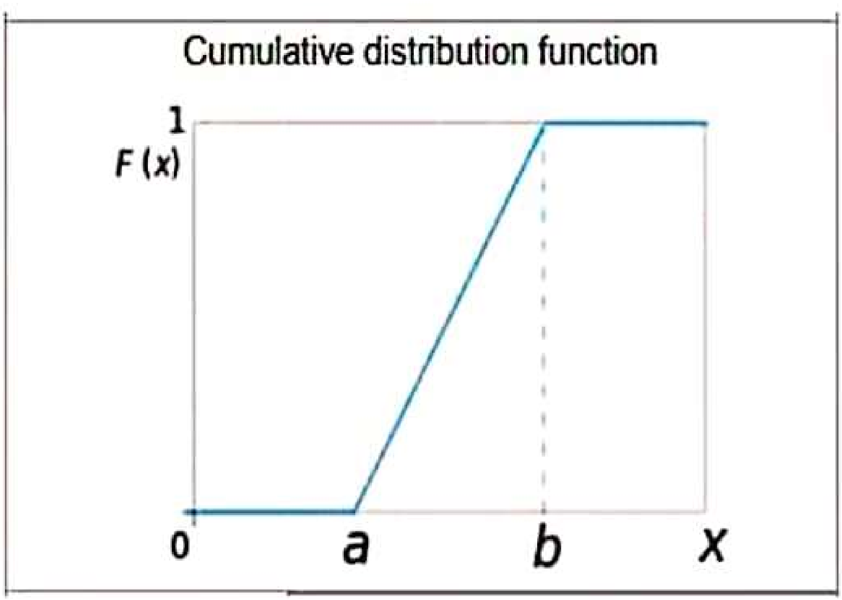
The Cumulative Distribution function of the continuous uniform distribution is:

$$F(x)=\begin{cases}0 & \text{for } x<a \\ \dfrac{x-a}{b-a} & \text{for } a\le x\le b, \\ 1 & \text{for } x>b\end{cases}$$

Derivation of the expression for the interval $a \le x \le b$:

$$F(x)=\int_a^x f(x)\,dx=\int_a^x \frac{1}{b-a}\,dx=\frac{1}{b-a}\int_a^x 1\,dx$$

$$=\frac{1}{b-a}[x]_a^x=\frac{1}{b-a}[x-a]=\frac{x-a}{b-a}$$



Why is it a straight line between a and b ?

$$F(x)=\frac{x-a}{b-a}=\frac{x}{b-a}-\frac{a}{b-a}$$

or

$$F(x)=\left(\frac{1}{b-a}\right)x+\left(-\frac{a}{b-a}\right)$$

which is of the form $y=mx+c$.

Proof of Variance:

By definition,

$$\operatorname{Var}(X) = E(X^2) - [E(X)]^2. \qquad 1$$

Now,

$$E(X^2) = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$

$$= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2 + ab + b^2)}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

Put in (1)

Therefore,

$$\operatorname{Var}(X) = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{a^2 + ab + b^2}{3} - \frac{a^2 + 2ab + b^2}{4} = \frac{4a^2 + 4ab + 4b^2 - 3a^2 - 6ab - 3b^2}{12}$$

$$= \frac{a^2 - 2ab + b^2}{12} = \frac{(b-a)^2}{12}.$$

✓ gmP ✓

02:5

0 PDF of the **Exponential Distribution** with mean= $\theta = 0.5, 1, 1.5$ = 0.6

- $f(x)=\left(\frac{1}{0.5}\right)e^{-\left(\frac{x}{0.5}\right)}, \quad 0<x<\infty$ → orange ✓
- $f(x)=e^{-(x)}, \quad 0<x<\infty$ = 1.0 → purple ✓
- $f(x)=\left(\frac{1}{1.5}\right)e^{-\left(\frac{x}{1.5}\right)}, \quad 0<x<\infty$ → skyblue ✓

= 1.6

= 0.6

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The Probability density function (pdf) of an exponential distribution is

$$f(x;\lambda)=\begin{cases}\lambda e^{-\lambda x} & x\geq 0,\ \lambda>0\\ 0 & x<0.\end{cases}$$

So it is obvious that is a **single-parameter** distribution.

Let $\theta=\frac{1}{\lambda}$

then,

mean $=\theta$

and

$$f(x;\theta)=\begin{cases}\frac{1}{\theta}e^{-\frac{x}{\theta}} & x\geq 0,\ \theta>0\\ 0 & x<0.\end{cases}$$

So it is obvious that is a **single-parameter** distribution.

First and foremost, let us consider the <u>shape</u> of the distribution:

PDF of the **Exponential Distribution** with mean= $\theta=0.5,\ 1,\ 1.5$

- $f(x)=\left(\frac{1}{0.5}\right)e^{-\left(\frac{x}{0.5}\right)},\quad 0<x<\infty$ → orange
- $f(x)=e^{-(x)},\quad 0<x<\infty$ → purple
- $f(x)=\left(\frac{1}{1.5}\right)e^{-\left(\frac{x}{1.5}\right)},\quad 0<x<\infty$ → skyblue

Next: CDF

The Cumulative distribution function (cdf) of an exponential distribution is

$$F(x;\lambda)=\begin{cases}1-e^{-\lambda x} & x\geq 0,\ \lambda>0\\ 0 & x<0.\end{cases}$$

The Cumulative distribution function (cdf) of an exponential distribution in terms of θ

$$F(x;\theta)=\begin{cases}1-e^{-\frac{x}{\theta}} & x\geq 0,\ \theta>0\\ 0 & x<0.\end{cases}$$

Proof:

$$F(x)=\int_{-\infty}^{x} f(x)dx=\int_{0}^{x}\lambda e^{-\lambda x}dx=-\int_{0}^{x}(-\lambda)e^{-\lambda x}dx$$

$$=-\left[e^{-\lambda x}\right]_{0}^{x}=-\left[e^{-\lambda x}-e^{0}\right]=-\left[e^{-\lambda x}-1\right]$$

or

$$F(x)=1-e^{-\lambda x},\ x\geq 0$$

Now, let us consider the <u>shape</u> of the CDF:

Topic No. 157

MGF of the Exponential Distribution

The integral is called the **gamma function**, and we write

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy$$

Properties of the Gamma Function

- If $\alpha = 1$, clearly (easy to verify through integration) $\Gamma(1) = \int_0^\infty e^{-y} dy = 1$
- If $\alpha > 1$, an integration by parts show that

$$\Gamma(\alpha) = (\alpha-1)\int_0^\infty y^{\alpha-2} e^{-y} dy = (\alpha-1)\Gamma(\alpha-1).$$

- Accordingly, if α is a positive **integer** greater than 1,

$$\Gamma(\alpha) = (\alpha-1)(\alpha-2)\cdots(3)(2)(1)\Gamma(1) = (\alpha-1)!.$$

- Since $\Gamma(1) = 1$, this suggests we take **0! =1.**
- It is important to note that **the value of a gamma function is always positive.**

The mean of this distribution is 16.

&

The variance of this distribution is 16.

Therefore,

The **coefficient of variation** of this distribution is

$$COV = \frac{\sigma}{\mu} = \frac{\sigma}{\mu} \times 100\% = \frac{\sqrt{\lambda}}{\lambda} \times 100\% = \frac{1}{\sqrt{16}} \times 100\%$$

$$= \frac{1}{4} \times 100\% = 25\%$$

It appears that, as λ increases, the coefficient of variation of the Poisson distribution decreases.

Stated a little more rigorously:

For a Poisson distribution with parameter λ, as λ tends to infinity, the coefficient of variation of the Poisson distribution also tends to 0.

Virtual University of Pakistan
Probability Distributions

by

$\lambda = 9$ ∴ 33.3%

Dr. Saleha Naghmi Habibullah

$h = 4$ ∴ 50%

Topic No. 145

Derivation

If it is known that, on the average, **two** minor accidents occur at a busy crossing --- per day, what is the probability that, **at the <u>most</u> two** minor accidents will occur during an entire week ?

Exp

Solution:

This is a Poisson process with Occurrence Rate = λ = 2 per day.

Also, one week = 7 days implies t=7

Hence

So that $\lambda t = 2 \times 7 = 14$

$$P(X=x)=\begin{cases}\dfrac{e^{-\lambda t}(\lambda t)^x}{x!}=\dfrac{e^{-14}(14)^x}{x!} & x=0,1,2,\ldots \\ 0 & elsewhere,\end{cases} \qquad (1)$$

Now, we require X<2.

So

$$P(X \le 2) = P(X=0 \text{ or } X=1 \text{ or } X=2)$$
$$= \frac{e^{-14}(14)^0}{0!} + \frac{e^{-14}(14)^1}{1!} + \frac{e^{-14}(14)^2}{2!}$$
$$= e^{-14} + 14e^{-14} + 98e^{-14} = 113e^{-14}$$
$$= 113/1202592.9591 = 0.00009396$$

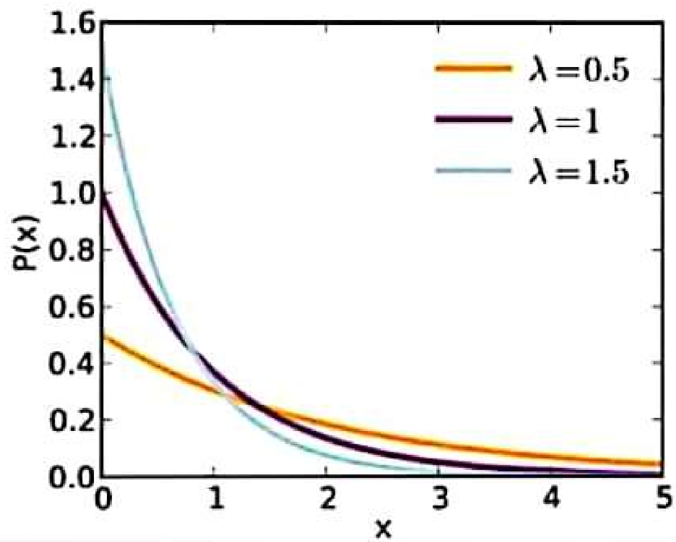
We can say that the probability that at the most 2 accidents will occur at this particular crossing in a week, this probability is almost 0, which is "Next to Impossible!"

the shape of the distribution:

PDF of the **Exponential Distribution**

with mean= $\theta = 0.5, 1, 1.5$

- $f(x) = \left(\frac{1}{0.5}\right) e^{-\left(\frac{x}{0.5}\right)}, \quad 0 < x < \infty$ → orange
- $f(x) = e^{-(x)}, \quad 0 < x < \infty$ → purple
- $f(x) = \left(\frac{1}{1.5}\right) e^{-\left(\frac{x}{1.5}\right)}, \quad 0 < x < \infty$ → skyblue



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Derivation of
Poisson Approximation to the Binomial distribution

To derive an approximation formula to the binomial distribution $b(x\,;\,n\,,p)$ when $n \to \infty$, $p \to 0$, and the product np remains **constant** we proceed :

An Example of the Poisson Approximation to the Binomial distribution

- To derive an approximation formula to the binomial distribution $b(x;n, p)$ where $n \to \infty$, $p \to 0$, and the product np remains **constant**, we proceed:

Example:

Suppose 8% of the tires manufactured at a particular plant are defective. To illustrate the use of the Poisson approximation for the binomial, the probability of obtaining exactly one defective tire from a sample of 20 using equation (5.18) is calculated as follows:

$$P(X=1) \equiv \frac{e^{-(20)(.08)}\left[(20)(.08)\right]^1}{1!} = \frac{e^{-1.6}\left[1.6\right]^1}{1!} = 0.3230$$

Had the true distribution, the binomial, been used instead of the approximation,

$$P(X=1) \equiv \binom{20}{1}(.08)^1(.92)^{19} = 0.3282$$

$$P(X < 9.46) = \frac{1}{4}\left(-2e^{-x/2}x - 4e^{-x/2}\right)_0^{9.46}$$

$$= \frac{1}{4}\left[\left(-2e^{-9.46/2}(9.46) - 4e^{-9.46/2}\right) - \left(-2e^{-0/2}.0 - 4e^{-0/2}\right)\right]$$

$$= \frac{1}{4}\left[(0.16700 - 0.003531) - \left(-0 - 4e^{0}\right)\right]$$

$$= \frac{1}{4}\left[(-0.20230) - (-4)\right] = \frac{1}{4}\left[(-0.20230) + (4)\right] = \frac{3.79770}{4} = 0.9494.$$

In other words, the probability that $X < 9.46$ is 94.94% almost 95 %.

Virtual University of Pakistan
Probability Distributions

by

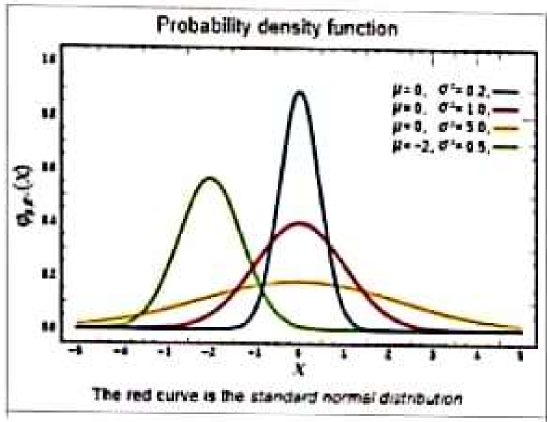
Dr. Saleha Naghmi Habibullah

Topic No. 161

MGF of Gamma Distribution

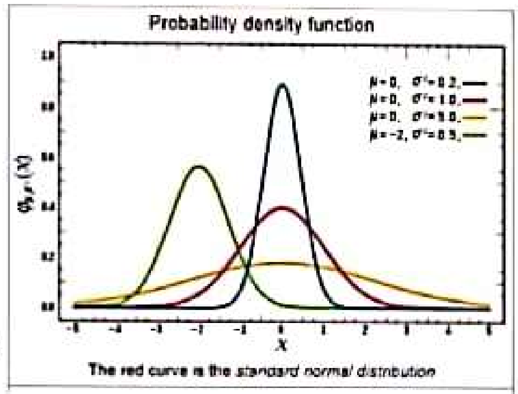
The graph points to the following **two basic properties** of the normal distribution:

1. The normal curve is **asymptotic** to the x-axis,



2. The maximum ordinate (i.e. the **modal ordinate**) of the normal distribution at $x=\mu$ is equal to $1/(\sigma\sqrt{2\pi})$

$$f(x)=\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty<x<\infty.$$



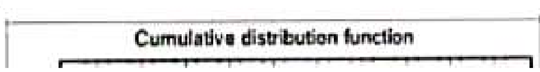
Cumulative Distribution Function

The formula for the **cumulative distribution function** of the normal distribution is

where
$$F(x)=\frac{1}{2}\left[1+\operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right] \quad \text{for } -\infty<x<\infty.$$

erf(z) is the "error function" defined by
$$\operatorname{erf}(z)\equiv\frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\,dt.$$

The shape of the **cumulative distribution function** of the normal distribution is **S-shaped**:



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$$f''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}\left[1 - e^{-(x-\mu)^2/2\sigma^2}\frac{(x-\mu)^2}{\sigma^2}\right]$$

Substituting $x = \mu$ in $f''(x)$, we see that $f''(x) < 0$.

Thus,

$x = \mu$ is the mode of the normal distribution.

Hence the median and mode are both equal to μ.

Since,

$$\text{mean} = \text{median} = \text{mode},$$

the normal distribution is symmetrical and unimodal.

- The **probability mass function** of this multinomial distribution is:

$$f(x_1, \cdots, x_k; n, p_1, \cdots, p_k) = \begin{cases} \dfrac{n!}{x_1!x_2!\cdots x_k!} p_1^{x_1} \times p_2^{x_2} \cdots \times p_k^{x_k}, & \text{when } \sum_{i=1}^{k} x_i = n \\ 0 & \text{otherwise.} \end{cases}$$

for non-negative integers $x_1, \ldots, x_k$.

This is the multinomial pmf of k-1 random variables $X_1, X_2, \ldots, X_{k-1}$ of the discrete type.

Properties:

The **mean** of the Multinomial distribution

Topic No. 165

Normal Distribution
(PDF, CDF and Shape of the distribution)

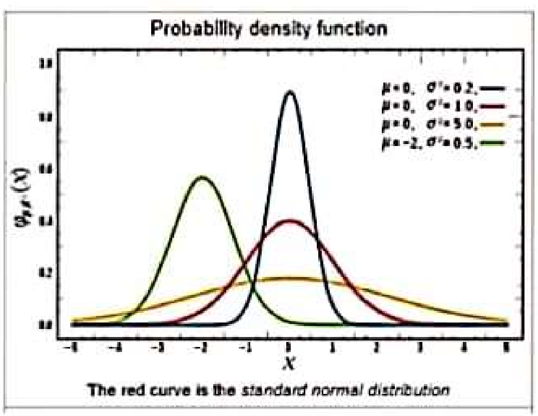
Definition:

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty.$$

- μ is the mean or expectation of the distribution (and also its median and mode),
- $\sigma > 0$ is the standard deviation, and
- σ^2 is the variance.

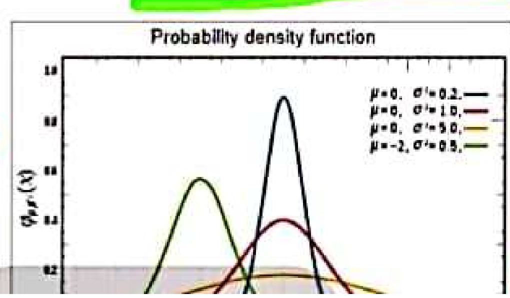
Shape of the distribution:



The red curve is the standard normal distribution

The graph points to the following **two basic properties** of the normal distribution:

1. The normal curve is **asymptotic** to the x-axis,



Properties:

The **mean** of the Multinomial distribution

$$E(X_i) = np_i$$

The **variance** of the Multinomial distribution

$$\text{Var}(X_i) = np_i(1 - p_i)$$

The **covariance** of the Multinomial distribution

$$\text{cov}(X_i, X_j) = -np_i p_j, \quad i \neq j$$

Note:

The binomial distribution is **a special case** of the multinomial distribution.

Geometric Distribution
(PMF and shape of the distribution)

(4) n, the number of trials is fixed in advance
and
the binomial distribution is given by

$$p_X(x)=\binom{n}{x}p^x(1-p)^{n-1},\ x=0,1,2,3,\ldots,n$$

When n=1,
there are two possible outcomes,
and p, the probability of success, is the only parameter of the Bernoulli distribution given by

$$p_X(x)=\binom{1}{x}p^x(1-p)^{1-x},\ x=0,1$$

or

$$p_X(x)=p^x(1-p)^{1-x},\ x=0,1$$

If we let X denote the number of trials **until the first success.**

Then, the **probability mass function** of X is:

$$p_X(x)=(1-p)^{x-1}p,\ x=1,2,3,\ldots$$

Example:

Suppose that a game consists of tossing a fair coin until we obtain the first head.

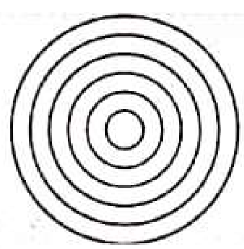
Then $p=P(H)=0.5$

And, in this case, the geometric distribution is given by

By the term **'contour'**, we mean the **set of points on the X_1X_2 floor** corresponding to which **the ordinates** are of **<u>equal height</u>.**

<u>Explanation</u>: Consider the bivariate normal distribution for which $\boldsymbol{\rho = 0}$ and $\sigma_1 = \sigma_2 = \sigma$.

In this case the **contours** of the distribution will be **concentric circles** because, against **each** of these **<u>circles</u>** (on the X_1X_2 floor), the ordinates are of **<u>equal height</u>.**



- The ordinates against <u>the outer circle</u> are <u>very short;</u>
- The ordinates against <u>the Second circle (inner circle)</u> are <u>a little taller</u> than the ones against the outer circle;
- The ordinates against <u>the Third (inner) circle</u> are <u>taller</u> than the ones against the Second circle;
- The ordinates against <u>the Fourth (inner) circle</u> are <u>taller</u> than the ones against the Third circle;

And so on.

The **circular contour** corresponding to a lower height is **wider** than the one corresponding to a greater height.

$f(x_1, x_2)$

PDF of the bivariate normal distribution for the case $\rho = 0$.

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\left\{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)\right\}}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-0^2}} e^{-\frac{1}{2(1-0^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2(0)\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1}} e^{-\frac{1}{2(1)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 0 + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)} = \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{1}{2}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{a^2} = 1 \Rightarrow (x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

and we **already** know that

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

is the equation of **<u>that</u> particular circle** with center at the point (c_1, c_2) and with radius a.

Hence, if we put $\sigma_1 = \sigma_2 = \sigma$ in equation (2), we obtain

$$\frac{(x_1 - \mu_1)^2}{k\sigma^2} + \frac{(x_2 - \mu_2)^2}{k\sigma^2} = 1$$

$$\Rightarrow (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 = k\sigma^2 \qquad ...(3)$$

which is the equation of a <u>circle</u>
with center at (μ_1, μ_2) and with radius $= k\sigma$.

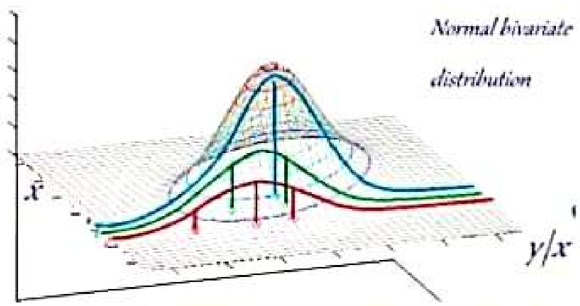
y

is a **particular** **value** of X, i.e. it is a **constant.**

Because of the above, we can see that:

As far as the **mean** of the random variable Y corresponding to any particular value of X is concerned,

$$\tilde{\mu}_y = \mu_y + (x - \mu_x)\left(\frac{\sigma_y}{\sigma_x}\right)\rho$$



https://www.researchgate.net/figure/Stage-wise-conditional-distributions-sitting-on-the-tree-Gaussian-case_fig11_[illegible]

represents the mean value of y against some **particular** **value** of x.

A Special Case of the Negative Binomial Distribution:

Geometric Distribution

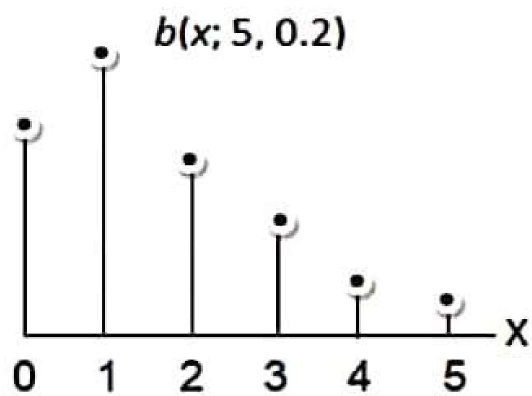
Topic No. 122

Shape

of the

Binomial distribution

- The **shape** of the binomial probability distribution depends on the values of the **two parameters** p and n.
- The sketches given indicate the influence of p and n on the shape of the distribution.



Topic No. 122

Shape
of the
Binomial distribution

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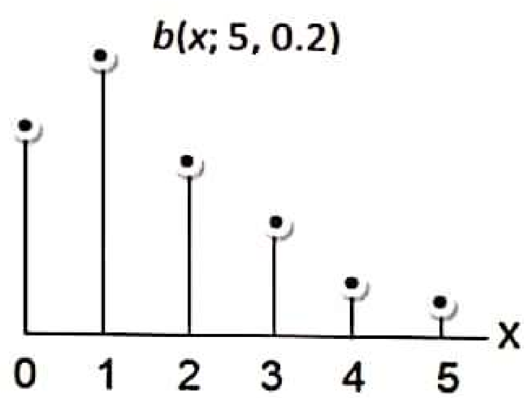


Topic No. 120

Binomial distribution

(Binomial experiment and PMF of the distribution)

Binomial Experiment

An experiment is called a **binomial experiment** if it possesses the following **four** properties:

1. The outcome of each trial may be classified into one of two categories, conventionally called **success (S)** and **Failure (F).**

   It is to be noted that the outcome of **interest** is called a **success** an the other, a **failure.**

2. The probability of success, denoted by p; remains **constant** for all trials.

3. The successive trials are all **independent.**

4. The experiment is repeated a **fixed** number of times, say n.

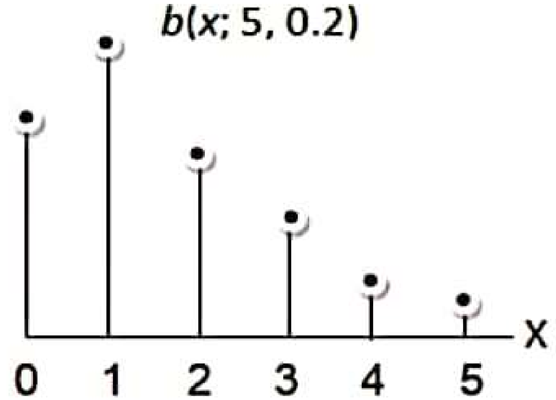
Topic No. 122

Shape

of the

Binomial distribution

- The **shape** of the binomial probability distribution depends on the values of the **two parameters** p and n.
- The sketches given indicate the influence of p and n on the shape of the distribution.



The **probability density function** of the β distribution is

$$f(x;\alpha,\beta)=\frac{1}{B(\alpha,\beta)}x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0\le x\le 1;\alpha,\beta>0$$

where,

$$B(\alpha,\beta)=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

and α and β are the **shape parameters** respectively.

About the Beta function B

- The Beta function B in the denominator plays the role of a "**normalizing constant**" which assures that the total area under the density curve equals 1.

- The Beta function is equal to a **ratio** of Gamma functions:

$$B(\alpha,\beta)=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

- Keeping in mind that for integers, $\Gamma(k) = (k-1)!$, one can do some checking and get an idea of what the shape might be.

Some Properties:

- **Mean** $=\mu$

$$= E(x) = \alpha/\alpha+\beta$$

- **Variance(x)**

$$= \alpha\beta \,/\, (\alpha+\beta)^2\,(\alpha+\beta+1)$$

Mode:

- If $\alpha > 1$ and $\beta > 1$, the peak of the density is in the interior of [0,1] and mode of the Beta distribution is

$$\textbf{mode} = \alpha-1/\alpha+\beta-2$$

If α or $\beta < 1$, the mode may be at an edge.

RECALL

the Mathematical Equation of An Ellipse

The equation of an ellipse with major axis parallel to the X_1-axis and minor axis parallel to the X_2-axis and having its center at (c_1,c_2) is as follows:

$$\frac{(x_1-c_1)^2}{a^2}+\frac{(x_2-c_2)^2}{b^2}=1 \qquad ...(1)$$

Next:

The Shape of the distribution when $\rho = 0$ AND $\sigma_1 = \sigma_2$

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1-c_1)^2}{a^2}+\frac{(x_2-c_2)^2}{a^2}=1 \Rightarrow (x_1-c_1)^2+(x_2-c_2)^2=a^2$$

and we **already** know that

$$(x_1-c_1)^2+(x_2-c_2)^2=a^2$$

is the equation of **that particular circle** with center at the point (c_1,c_2) and with radius a.

Hence, if we put $\sigma_1 = \sigma_2 = \sigma$ in equation (2), we obtain

$$\frac{(x_1-\mu_1)^2}{\;}+\frac{(x_2-\mu_2)^2}{\;}=1$$

$$\rho = 0 \text{ AND } \sigma_1 = \sigma_2$$

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{a^2} = 1 \Rightarrow (x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

and we **already** know that

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

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Hence, if we put $\sigma_1 = \sigma_2 = \sigma$ in equation (2), we obtain

$$\frac{(x_1 - \mu_1)^2}{k\sigma^2} + \frac{(x_2 - \mu_2)^2}{k\sigma^2} = 1$$

$$\Rightarrow (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 = k\sigma^2 \qquad \text{...(3)}$$

which is the equation of a circle
with center at (μ_1, μ_2) and with radius $= k\sigma$.

Note:

The binomial distribution is **a special case** of the multinomial distribution.

Next:

A circle can be regarded as

a special case

of an **ELLIPSE.**

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{t_1x_1}e^{t_2x_2} f_1(x_1)f_2(x_2)\,dx_1dx_2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{t_1x_1+t_2x_2} f_1(x_1)f_2(x_2)\,dx_1dx_2.$$

We are given that,

$$M(t_1,t_2) = M(t_1,0)M(0,t_2);$$

Thus

$$M(t_1,t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{t_1x_1+t_2x_2} f_1(x_1)f_2(x_2)\,dx_1dx_2 \qquad (1)$$

But, by definition, $M(t_1,t_2)$ is the mgf of X_1 and X_2

i.e.

$$M(t_1,t_2) = E\left(e^{t_1X_1+t_2X_2}\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{t_1x_1+t_2x_2} f(x_1,x_2)\,dx_1dx_2 \qquad (2)$$

The uniqueness of the mgf implies that the two distributions of probability that are described by $f_1(x_1)f_2(x_2)$ and $f(x_1,x_2)$ are the same.

Thus

$$f(x_1,x_2) \equiv f_1(x_1)f_2(x_2).$$

That is, if $M(t_1,t_2) = M(t_1,0)M(0,t_2)$, then X_1 and X_2 are independent.

Binomial Distribution

Let the random variable X denote the number of successes in n trials --- when a binomial experiment is performed.

Then the PMF of X is given by

$$p(x)=\begin{cases}\binom{n}{x}p^{x}(1-p)^{n-x} & x=0,1,2,\ldots,n \\ 0 & \textit{elsewhere.}\end{cases} \qquad (1)$$

where p is the probability of success in each trial and n is the number of trials.

We can now write these probabilities in the form of a probability table as below:

x	0	1	2	3	4	5
$P(X=x)$	0.0954	0.2861	0.3233	0.2060	0.0618	0.0074

Binomial Theorem

1004

$$(q+p)^n = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x}$$

- When $p = 1/2$, the distribution is always symmetrical.

In general:

- As n, the number of trials, increases to ∞, $\beta_1 \rightarrow 0$ and $\beta_2 \rightarrow 3$.

- Hence for large n, the binomial probability distribution is symmetrical and mesokurtic.

We are given that,

$$M(t_1,t_2) = M(t_1,0)M(0,t_2);$$

Thus

$$M(t_1,t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{t_1x_1+t_2x_2} f_1(x_1)f_2(x_2)\,dx_1dx_2 \qquad (1)$$

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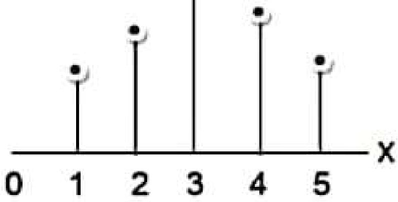
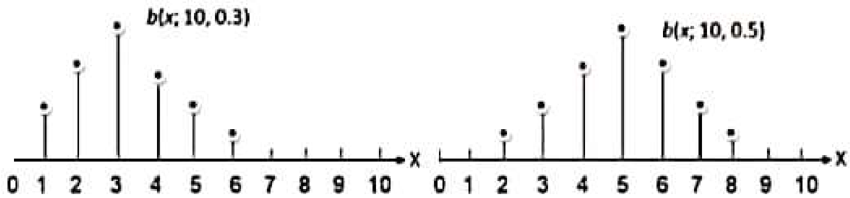
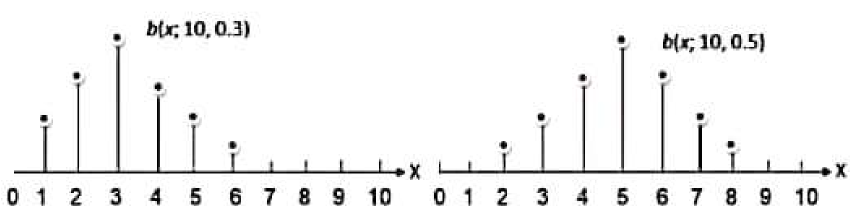




- Thus, we observe that, when $p < ½$, the distribution is positively skewed and when $p > ½$, the distribution becomes negatively skewed.

1012



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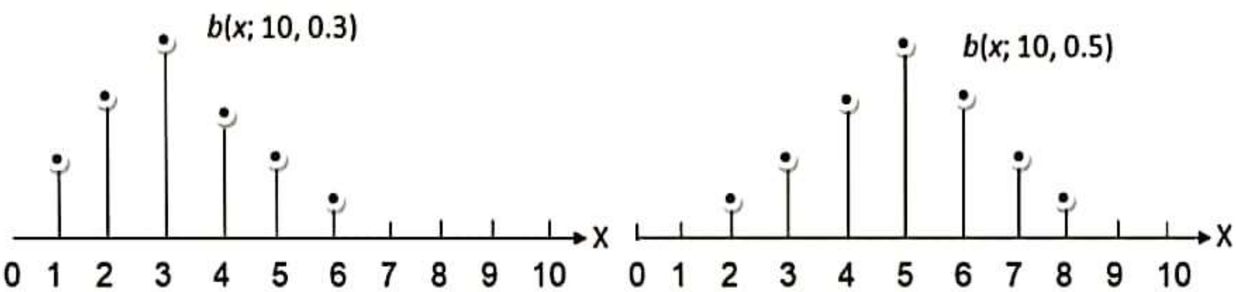

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1012

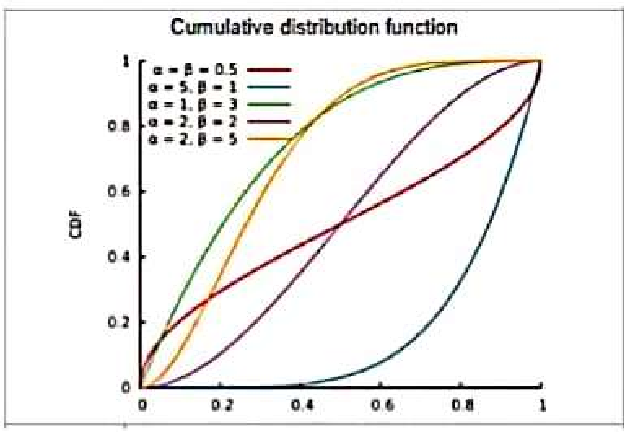
Cumulative distribution Function

The formula for the cumulative distribution function of the beta distribution is also called the **incomplete β function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha,\beta) = \frac{\int_0^x t^{\alpha-1}(1-t)^{\beta-1}}{B(\alpha,\beta)}, \qquad 0 \le x \le 1; \alpha, \beta > 0$$

where B is the beta function defined above.

Shape of the CDF:



Some Properties:

- **Mean** $= \mu$

$$= E(x) = \alpha/\alpha+\beta$$

- **Variance(x)**

$$= \alpha\beta \,/\, (\alpha+\beta)^2\, (\alpha+\beta+1)$$

Definition

A pair of random variables X_1 and X_2 have a bivariate normal distribution and they are referred to as **jointly normally distributed** random variables **if and only if** their joint probability density is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right\}}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_2 < \infty$,

where $\mu_1 \in \mathbb{R}, \mu_2 \in \mathbb{R}, \sigma_1 > 0, \sigma_2 > 0$, and $-1 \le \rho \le 1$.

This is referred to as the $N\left(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho\right)$ distribution.

This joint <u>p.d.f.</u>, has a <u>bell</u>-type shape.

Bivariate Normal Distribution with $\rho = 0$ and $\sigma_1 = \sigma_2 = \sigma$

- The standard normal distribution is a **special** case of the normal distribution .

What is meant by the 'X_1X_2 floor'?

Consider the three-dimensional space for example the room in which you are sitting right now.

•Now focus on the floor of the room.

•One 'edge' of the floor will be called the X_1 axis

•The other 'edge' of the floor will be called the X_2 axis

And, as such, the floor will be called the 'X_1X_2 floor'
(actually it is the X_1X_2-plane)

Next:

The concept of

Contours

By the term **'contour'**, we mean the **set of points on the X_1X_2 floor** corresponding to which **the ordinates** are of **<u>equal height</u>.**

<u>Explanation</u>: Consider the bivariate normal distribution for which $\rho = 0$ and $\sigma_1 = \sigma_2 = \sigma$.

In this case the **contours** of the distribution will be

$$1-\left(\frac{2}{3}\right)^n > 0.80$$

$$-\left(\frac{2}{3}\right)^n < -0.80+1$$

$$\left(\frac{2}{3}\right)^n < 0.20$$

Taking logs:-

$$\ln(0.20) > \ln(2/3)^n$$

using log property

$$\ln(0.20) > n\ln\left(\frac{2}{3}\right)$$

$$-1.6094 > n\ln(0.6666)$$

$$-1.6094 > n(-0.4055)$$

Multiplying by -1 on b.s.

$$1.6094 < 0.4055n$$

$$n > \frac{1.6094}{0.4055}$$

$$n > 3.97$$

n must be whole no. so

$$\boxed{n=4}$$

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Recognizing the values of the parameters of a Binomial distribution through the expression of its MGF
(explained through an example)

Example:

If the mgf of a random variable X is

$$M(t)=\left(\frac{2}{3}+\frac{1}{3}e^t\right)^5,$$

Then, X has a binomial distribution with $n = 5$ and $p = 1/3$;

---

that is, the **pmf** of X is

$$p(x)=\begin{cases}\binom{5}{x}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{5-x} & x=0,1,2,\ldots,5 \\ 0 & \textit{elsewhere.}\end{cases}$$

---

and

$$\mu=np=\frac{5}{3}\ \ \textit{and}\ \ \sigma^2=np(1-p)=\frac{10}{9}.$$

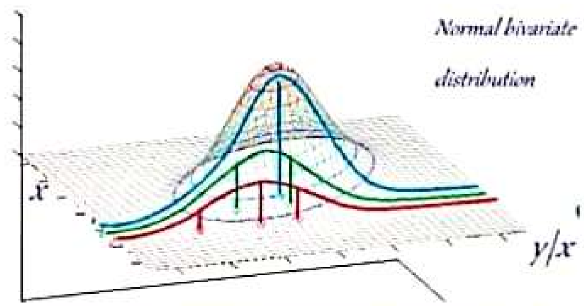
y

is a **particular** **value** of X, i.e. it is a **constant.**

Because of the above, we can see that:

As far as the **mean** of the random variable Y corresponding to any particular value of X is concerned,

$$\tilde{\mu}_y = \mu_y + (x - \mu_x)\left(\frac{\sigma_y}{\sigma_x}\right)\rho$$



https://www.researchgate.net/figure/Stage-wise-conditional-distributions-sitting-on-the-tree-Gaussian-case_fig11_299488305

represents the mean value of y against some **particular** **value** of x.

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{a^2} = 1 \Rightarrow (x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

and we **already** know that

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

is the equation of **<u>that</u> particular circle** with center at the point (c_1,c_2) and with radius a.

Hence, if we put $\sigma_1 = \sigma_2 = \sigma$ in equation (2), we obtain

$$\frac{(x_1 - \mu_1)^2}{k\sigma^2} + \frac{(x_2 - \mu_2)^2}{k\sigma^2} = 1$$

$$\Rightarrow (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 = k\sigma^2 \qquad ...(3)$$

which is the equation of a <u>circle</u>

with center at (μ_1, μ_2) and with radius $= k\sigma$.

RECALL

the Mathematical Equation of An Ellipse

The equation of an ellipse with major axis parallel to the X_1-axis and minor axis parallel to the X_2-axis and having its center at (c_1,c_2) is as follows:

$$\frac{(x_1-c_1)^2}{a^2}+\frac{(x_2-c_2)^2}{b^2}=1 \qquad ...(1)$$

Next:

The Shape of the distribution

when

$\rho = 0$ AND $\sigma_1 = \sigma_2$

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1-c_1)^2}{a^2}+\frac{(x_2-c_2)^2}{a^2}=1 \Rightarrow (x_1-c_1)^2+(x_2-c_2)^2=a^2$$

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Hence, if we put $\sigma_1 = \sigma_2 = \sigma$ in equation (2), we obtain

$$\frac{(x_1-\mu_1)^2}{\quad}+\frac{(x_2-\mu_2)^2}{\quad}=1$$

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which is the equation of a <u>circle</u>

with center at (μ_1, μ_2) and with radius $= k\sigma$.

PDF of the bivariate normal distribution for the case $\rho = 0$.

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\left\{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)\right\}}$$

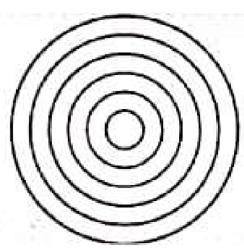
$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-0^2}} e^{-\frac{1}{2(1-0^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2(0)\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1}} e^{-\frac{1}{2(1)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 0 + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)} = \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{1}{2}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

By the term **'contour'**, we mean the **set of points on the X_1X_2 floor** corresponding to which **the ordinates** are of **equal height.**

Explanation: Consider the **bivariate** normal distribution for which $\boldsymbol{\rho} = \mathbf{0}$ and $\sigma_1 = \sigma_2 = \sigma$.

In this case the **contours** of the distribution will be **concentric circles** because, against **each** of these **circles** (on the X_1X_2 floor), the ordinates are of **equal height.**



- The ordinates against the outer circle are very short;
- The ordinates against the Second circle (inner circle) are a little taller than the ones against the outer circle;
- The ordinates against the Third (inner) circle are taller than the ones against the Second circle;
- The ordinates against the Fourth (inner) circle are taller than the ones against the Third circle;

And so on.

The **circular contour** corresponding to a lower height is **wider** than the one corresponding to a greater height.

$f(x_1, x_2)$

What is meant by the 'X_1X_2 floor'?

Consider the three-dimensional space for example the room in which you are sitting right now.

•Now focus on the floor of the room.

•One 'edge' of the floor will be called the X_1 axis

•The other 'edge' of the floor will be called the X_2 axis

And, as such, the floor will be called the 'X_1X_2 floor'

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By the term **'contour'**, we mean the **set of points on the X_1X_2 floor** corresponding to which **the ordinates** are of **equal height.**

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for $-\infty < x_1 < \infty$ and $-\infty < x_2 < \infty$,

where $\mu_1 \in \square, \mu_2 \in \square, \sigma_1 > 0, \sigma_2 > 0$, and $-1 \le \rho \le 1$.

This is referred to as the $N\left(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho\right)$ distribution.

This joint p.d.f., has a bell-type shape.

Bivariate Normal Distribution with $\rho = 0$ and $\sigma_1 = \sigma_2 = \sigma$

The **probability density function** of the β distribution is

$$f(x;\alpha,\beta)=\frac{1}{B(\alpha,\beta)}x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0\le x\le 1;\alpha,\beta>0$$

where,

$$B(\alpha,\beta)=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

and α and β are the **shape parameters** respectively.

About the Beta function B

- The Beta function B in the denominator plays the role of a "**normalizing constant**" which assures that the total area under the density curve equals 1.

- The Beta function is equal to a **ratio** of Gamma functions:

$$B(\alpha,\beta)=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

- Keeping in mind that for integers, $\Gamma(k) = (k-1)!$, one can do some checking and get an idea of what the shape might be.

Cumulative distribution Function

The formula for the cumulative distribution function of the beta distribution is also called the **incomplete β function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha,\beta) = \frac{\int_0^x t^{\alpha-1}(1-t)^{\beta-1}}{B(\alpha,\beta)}, \qquad 0 \le x \le 1; \alpha,\beta > 0$$

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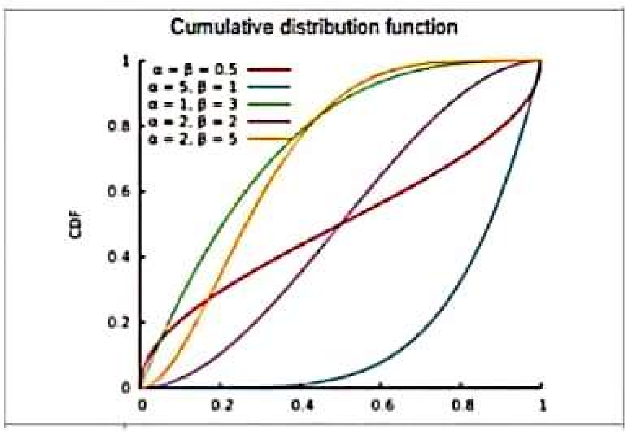
Shape of the CDF:



Some Properties:

- **Mean** $=\mu$

$$= E(x) = \alpha/\alpha+\beta$$

- **Variance(x)**

$$= \alpha\beta / (\alpha+\beta)^2 (\alpha+\beta+1)$$

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- **Mean** $= \mu$

$$= E(x) = \alpha/\alpha+\beta$$

- **Variance(x)**

$$= \alpha\beta \,/\, (\alpha+\beta)^2\,(\alpha+\beta+1)$$

Mode:

- If $\alpha > 1$and $\beta > 1$, the peak of the density is in the interior of $[0,1]$ and mode of the Beta distribution is

$$\textbf{mode} = \alpha-1/\alpha+\beta-2$$

If α or $\beta < 1$, the mode may be at an edge.

Subjectives:-

(1)

Write the formula for mean of multinomial dist. (2 Marks)

(2) Write two basic properties of normal distribution. (2)

(3) If $P = 0.2$ then find mean of Geometric distribution (2)

(4) Write the P.d.f of binomial distribution. (2) Marks

(5) Write the formula for variance of uniform distribution. (3 Marks

(6) When does two variable X and Y be independent? (2) Marks.

(7) If Pdf is

$$f(x,y) = \begin{cases} \frac{1}{8}(6-x-y), & 0 \le x \le 2, \\ & 2 \le y \le 4 \\ 0 & \text{elsewhere} \end{cases}$$

find marginal probability $g(x)$ (3 Marks)

If mean and variance equals to 4 then find Cov(X, Y). (5)

The probability that a planted radish seed is 0.70. A gardner plants nine seeds what is the probability that out of the nine seeds, at least 8 will germinate? (5)

If P = ? n = 5 and 'exactly one success required then use binomial distribution to find P(X=1)?

3 Marks

Write the formula for geometri distribution. 2 Marks.

Write formula for joint bivariate normal distribution (2 Marks)

Derive the conditional joint Probability for bivariate uniform distribution. (5 Marks)

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Formula of variance ?

obtained the 1^{st} and 2^{nd} cumlant normal distribution

write the mgf function and multinomal distribution

Possion distribution of binomal distribution

X, Y is bivartae then marginal PDF is of x.

Probility densitt function of st norml dstn

① Obtain the second Raw moments of the normal Distribution form its MGf.

② Given $f(x,y) = \begin{cases} 2 & 0 < x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$

③ Find $f(y|x)$

Suppos distribution larg western city i

white	Black	Hispanic
65%	20%	15%

Find twele members from city in such way that each resident has equal Probibity.

④ Prove $E(X) = \frac{a+b}{2}$ by using uniform continous distribution?

⑤ For Binomial distribution $b(x; 5, 1/3)$ find the $P(\alpha = c)$?

⑥ obtain First and 2^{nd} cumulant of the normal distribution ?

Sta642
Quiz k liay
All distributions (parameters
,Mean variance formulas) are very important
Mostly, hyper geometric distribution ,
bivariate, and geometric distribution me se
ziada quiz aty hain ye 3 aiss topics hain jin
me se sab k quiz aye hoay.

Or apko pata hona chahiay (pmf pdf CDF) kis
distribution k liay Kya use hota.
100% quiz✅✅
10:27 am

Date: / / Sta642

Q: if $P = 0.2$ find variance
of Bionomial Distribution. (2M)

Q: I Mean of Poisson Distribution
is 3 then what will be
the value of its variance? (3/2) شاید

Q: Derivation of Mean of
normal distribution. (5M)

random variable $X \sim U(1,n)$. (5)

→ Obtain the 2nd Raw Moments of the Normal Distribution from its MGF.? (3).

→ Define the role of normalizing constant in Beta distribution? (2)

→ $f(x,y)=\begin{cases} 2 & 0<x<y, 0<y<1 \\ 0 & \end{cases}$ Find marginal distribution of y. (2)

→ Which is the condition of derivative that is used to obtain the Point of inflection.? (2)

→ Formula of Negative Binomial Distribution? (2)

→ Write the any three Properties of Gamma Function.? (3)

→ If $P=0.2$ and $r=2$ then find the Mean of -ve Binomial Distribution? (3)

→ If conditional distribution is given by $f(x|y) = \frac{f(x,y)}{f(y)} = \frac{2}{2y} = \frac{1}{y}, 0<x<y$ then find the conditional Expectation of X^2 given $Y=y$ ie $E(X^2|Y=y)$ (3).

→ In Bivariate Normal distribution, what is meant by the $X_1 X_2$ floor? (3)

→ Suppose 8% of the Tires manufactured at a Particular plant are defective. We are interested to find the Probability of obtaining exactly one defective tire from a sample of 20. (10)

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

65 20 15 45
5 1 10
55

05

Suppose that the racial/ethnic distribution in a very large city

White	Black	Hispanic
65%	20%	15%

Jury of twelve members is to be chosen from it city in such a way that each resident has an equal probability

1) Derive mean of normal distribution?
2) Derive mean of standard distribution?
3) Malick bhai wali file ka sabse pehla question.
4) 20% black 80% green wala tha.

10

16-03-22 Stes 642 5:00 Am

3) Y if $P(0, \frac{1}{16}) = 1 - P(y \geq 1)$ $\Lambda = \frac{3}{4}$

find the smallest value of $P(y \geq 1) \geq 0.88$

say the ak sort Question

1

) $E(z) = \frac{a+b}{2}$ ye prove the 5 marks

3) Ak mcq bivariate uniform distribution ki shape

Konsi hoti U shape bell shape Rectangular

4) $\beta_2 =$ formula in greater than 3

then shape in ~~...~~ ?

5) Rectangle ka dusra name ?

$E(z)$ in equal ___?___ if x and y are independent ?

6) odd order moment ka mean ~~the~~ sh 0, 1, -1

7) even order moment 0, 1, -1, 0

8) Semi interquartile range ~~3 Marks~~ find karna th

r

$$f(x_1, x_2) \begin{cases} x_1 x_2 & 0 < x_1, x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the marginal probability of $P_x(x)$?

Uski koi file tu nahi hy but quiz conceptual thy like koi b distribution ly li gae uska mean kia variance kia hy pdf mgf btao.. Ya pdf dy dia r pocha yeh kiska hy... Interval پوچھا giya hy... Or k normal distribution kis distribution ka special case hy r shape parameters knsy hain... Or yeh k koi b distribution dy di gae usk parameter pochy gaey hain... R raw moment kia.. Uniform distribution ka dosra nam kia hy... Bivariate ko kia bola jata hy... Joind probability mein x y independent kb hoty hain.. Row=0 ho tu kia conditions fullfill hoti hy 🤔 or odd number ki value kia hoti hy.. Mean variance kis distribution main equal hoty hain mean standard deviation eqal hun tu knc distribution hoti hy... Covariance ka formula.. Marginal interval kaisy separate hoti hy.. Or first cumulants کیا hy 2nd cumulants kia hy higher cumulant kisk equal hoty hain
Agar beta constant ho tu alpa ki skewness kia hogi...
Agar p>1/2 ho ty kia hoga p<1/2 ho tu p=1/2 ho tu or agar p approach kary

Agar beta constant ho tu alpa ki skewness kia hogi...
Agar p>1/2 ho ty kia hoga p<1/2 ho tu p=1/2 ho tu or agar p approach kary zero ko tu kia hota hy... 😤 Beta =0 ho tu shape kia hogi.... Normal curve asymptotic hoti hy to which axis😑
Mode kis distribution main find hota hy... Kisi b distribution main sigma² kisy denote karta hy... Mew kis ki representation hy🌝

Mujhy yahi apny yaad hain... I think sary btaa diey hain
May be it's helpfull for you👆
#Billi🤭

Sta Ka paper mostly Ain 2 files sy Aya hai mujhy Zia formula type Aya hai hyper geometric formula mean variance properties unki. 4 mcqs past files sy. Baqi Ain files sy bny hoiy thy Ain ko bs Achi tarh tyar kr ly.achy number sy pass ho jaiy gy

Sta642

Mostly from files

Ik question tha obtain even moments of normal distribution

Quiz main shape of curve b thy or formulas b thy distribution k

M(t) of binomial distribution

Subjective file sa a rahi ha.

Mean variance ,beta gama distribution of formulas achy sa karny ha.

Or sari distribution k parameters or pdf achy sa karny ha.

Bivariate normal distribution k topic ko achy sau dackna mere papr m us m say quiz short or long teno m aya tha

STA642 Time 2:30

Mcq from handouts

* Write PDF of binomial Distribution
* Write PDF of exponential Distribution IF $\theta = 1.5$
* What are Three properties of Beta Distribution
* Covariane of Possion Distribution if mean is 4
* Write Mean of Multinomial distribution
  and variance

6:00 AM

MTH641 S KHAN AC...
Abas, Baber Mande B, Dosra...

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Sta642⬆️⬆️ Mega +2 quiz file kr or 2 subjective file achy se kry paper acha ho ga

2:12 AM

Stylish 2nd Smester

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Sta642 today's paper
2 marks
1. Mean likhna tha dicrete ka
2. Mean likhna tha multinomial ka
3. 2 Real life examples btani thi biavriate ki
4. Mean given tha poisson ka standard deviation find krni thi
3 marks
1. Conditions btani thi jin mai poisson apply hou
2. Module 108 wali example ka ak part tha
3. Cdf likhna normal distribution ka
4. X1X2 floor
5 marks
1. Discrete ka variance drive krna tha
2. Continuous distribution ka mean
3. Minor accident wali example
4. Normal ke pdf ka derivative le ke simplify krna tha

2:12 AM

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Message

6:01 AM

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Abas, Baber Mande B, Dosra...

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Mcq 5,6 mids se thy baqi handouts se thy
2:12 AM

Stylish 2nd Smester
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Distribution ke parameters shapes ranges mean yh sb mcq mai tha
2:12 AM

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Standard deviation or mean kis distribution ka same hota ha and variance or mean kis distribution ka same hota ha
2:12 AM

Stylish 2nd Smester
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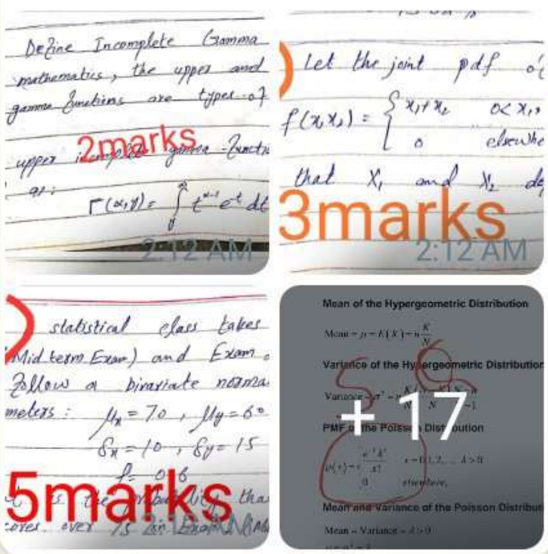

Current papers 2022

Message

6:00 AM

MTH641 S KHAN AC...
Abas, Baber Mande B, Dosra...

Stylish 2nd Smester
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Is file me sy aaj mery 20 sy 25 mcqs aaye hen.. **STA642**... 17 March 2:30pm
2:12 AM

Stylish 2nd Smester
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Subjective file sa a rahi ha.
Mean variance ,beta gama distribution of formulas achy sa karny ha.
Or sari distribution k parameters or pdf achy sa karny ha.
2:12 AM

Stylish 2nd Smester
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4 say 5 quiz ise topic pr thy
2:12 AM

Stylish 2nd Smester
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Bivariate normal distribution k topic ko achy sau dackna mere papr m us m say quiz short or long teno m aya tha
2:12 AM

Stylish 2nd Smester
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14 March ko mere 25+ mcqs isi file se ho gaye the. Aur 5 short bhi isi file se huye.
2:12 AM

Stylish 2nd Smester

Message

6:00 AM

MTH641 S KHAN AC...

Abas, Baber Mande B, Dosra...

Stylish 2nd Smester

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Bivariate normal distribution k topic ko achy sau dackna mere papr m us m say quiz short or long teno m aya tha

2:12 AM

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14 March ko mere 25+ mcqs isi file se ho gaye the. Aur 5 short bhi isi file se huye.

2:12 AM

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Aur 3 long is file se aaye. Mainne is file ko easy le liya tha ache se nai ki but phir bhi 2 long is file se huye.

2:12 AM

Stylish 2nd Smester

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1) Derive mean of normal distribution?
2) Derive mean of standard distribution?
3) Malick bhai wali file ka sabse pehla question.
4) 20% black 80% green wala tha.

2:12 AM

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Sta642⬆️⬆️ Mega +2 quiz file kr

Message

Sta642

1) Write formula for binomial distribution

2) The probability that a planted radish seed germinates is 0.80. A gardener plants nine seeds. What is the average number of seeds of gardener could expect to germinate.

3) The name of 5 men and 5 women are written on slips of paper and placed in a hat four names are drawn. What is the probability that 2 are men and 2 are women. 5

4 Derivation of the variance of the hypergeometric distribution. 5

5) Write pdf of poisson distribution

7) suppose 80% of the tires manufactured at a particular plant are defective. To illustrate the use of the poission approximation for the bionomial the probability of obtaining exactly one defective from a sample of 20. (5)

i) write mgf of gamma distribution.

9) write first and second commulants of normal distribution.

Sta642
Mostly from files
Ik question tha obtain even moments of normal distribution

Quiz main shape of curve b thy or formulas b thy distribution k

M(t) of binomial distribution

5:03 PM

Sta642
Quiz k liay
All distributions (parameters ,Mean variance formulas) are very important
Mostly, hyper geometric distribution , bivariate, and geometric distribution me se ziada quiz aty hain ye 3 aiss topics hain jin me se sab k quiz aye hoay.

Or apko pata hona chahiay (pmf pdf CDF) kis distribution k liay Kya use hota.
100% quiz✅✅

Sta642
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Ik question tha obtain even moments of normal distribution

Quiz main shape of curve b thy or formulas b thy distribution k

M(t) of binomial distribution 10:16 PM

+92 304 1350985 ~Ameen Ullah Khan
Forwarded
Pmf of multinomial 10:16 PM

+92 304 1350985 ~Ameen Ullah Khan
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Show that CDF of an exponential distrionbution is

(CDF given tha) 10:16 PM

+92 304 1350985 ~Ameen Ullah Khan
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1st derivative of cumulant 10:17 PM

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Forwarded
define the role of normalizing constant in binomial distribution

Virtual University

Rough Sheet

FALL/SPRING MID/FINAL TERM 20_ _ (Year) Date: ___/___/_____ Superintendent Signature:______

Campus Code: 11am Everblue Student Signature:______

Course Code: STA642 30/3/22 Student ID:______

1) Find/Simplify the first derivative of the normal distribution. (5)

2) Derive the formula for the MEAN of the Uniform Discrete random variable $X \sim U(1, n)$ (5)

3) Prove a Beta distribution with parameters $\alpha = 1, \beta = 1$ is a Uniform distribution on the interval $[0, 1]$. (5)

4) The name of 5 men and 5 women are written on slips of paper and placed in a hat. Four names are drawn. What is the probability that 2 men and 2 women? (Use Hypergeometric distribution). (5)

5) Write down two properties of a density function of normal distribution? (3)

6) If $P = 0.2$ then find the MEAN of Geometric distribution (3)

7) A Binomial experiments must fulfill few conditions. Write down any three of these conditions (3)

8) Derive the CV of exponential distribution? (3)

9) Give two real-life example of bivariate normal distribution (2)

10) In general, what is the impact of large 'n' on the shape of the binomial probability distribution? (2)

11) Write the moment generating function of the Gamma distribution (2)

12) Write down the formula of variance of the Hypergeometric distribution? (2)

Note: Submit this sheet to Superintendent, before leaving the Examination Cent...